PCOS ULTRASOUND CLASSIFICATION — MiT-B2 + MobileNet: EVERY RUN

Every result printed in the notebooks of MIT__MobileNet.zip (4 notebooks): 11 runs, including repeated runs of the same model + classifier. All of these were single exploratory runs (no fixed seed controlling the network training, or one seed without a repeat run) and are reported for completeness. Test set: 1,468 images (1,052 Appears abnormal, 416 Appears normal).

1. Overall Comparison — Best to Worst by Test Accuracy

Rank	Model	Classifier / run	Test Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	Source notebook (.ipynb)
1	MiT-B2 + MobileNetV2	Gaussian Naive Bayes (more epochs)	92.85	92.93	92.85	92.88	MIT more epochs.ipynb (cell 30)
2	MiT-B2 + MobileNetV2	Gaussian Naive Bayes	92.71	92.81	92.71	92.75	MIT + MobileNet +NB.ipynb (cell 35)
3	MiT-B2 + MobileNetV2	Gaussian Naive Bayes	92.64	92.77	92.64	92.69	MIT+ MobileNet+ ML.ipynb (cell 30)
4	MiT-B2 + MobileNetV2	KNN (more epochs)	88.96	89.24	88.96	88.39	MIT more epochs.ipynb (cell 30)
5	MiT-B2 + MobileNetV2	KNN	88.49	88.71	88.49	87.88	MIT+ MobileNet+ ML.ipynb (cell 30)
6	MiT-B2 + FPN	Random Forest	87.53	88.34	87.53	86.57	MIT + RF and MIT only.ipynb (cell 33)
7	MiT-B2 + MobileNetV2	Random Forest	87.06	87.94	87.06	85.99	MIT+ MobileNet+ ML.ipynb (cell 30)
8	MiT-B2 + MobileNetV2	Random Forest (more epochs)	86.85	87.56	86.85	85.82	MIT more epochs.ipynb (cell 30)
9	MiT-B2 + MobileNetV2	AdaBoost	86.51	87.48	86.51	85.32	MIT+ MobileNet+ ML.ipynb (cell 30)
10	MiT-B2 + MobileNetV2	AdaBoost (more epochs)	86.31	86.98	86.31	85.19	MIT more epochs.ipynb (cell 30)
11	MiT-B2 + FPN	Direct (fine-tuned CNN)	86.24	86.87	86.24	85.13	MIT + RF and MIT only.ipynb (cell 32)

- 11 runs in total; none is merged or dropped, so the same model + classifier can appear several times (different notebooks, or repeated runs inside one notebook).
- Precision, Recall and F1-score are class-weighted averages.
- Section 3 lists all further notes.

2. Results, Confusion Matrices and Graphs — Table Order

1. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes (more epochs)

Source notebook: MIT more epochs.ipynb (inside MIT + MobileNet.zip)

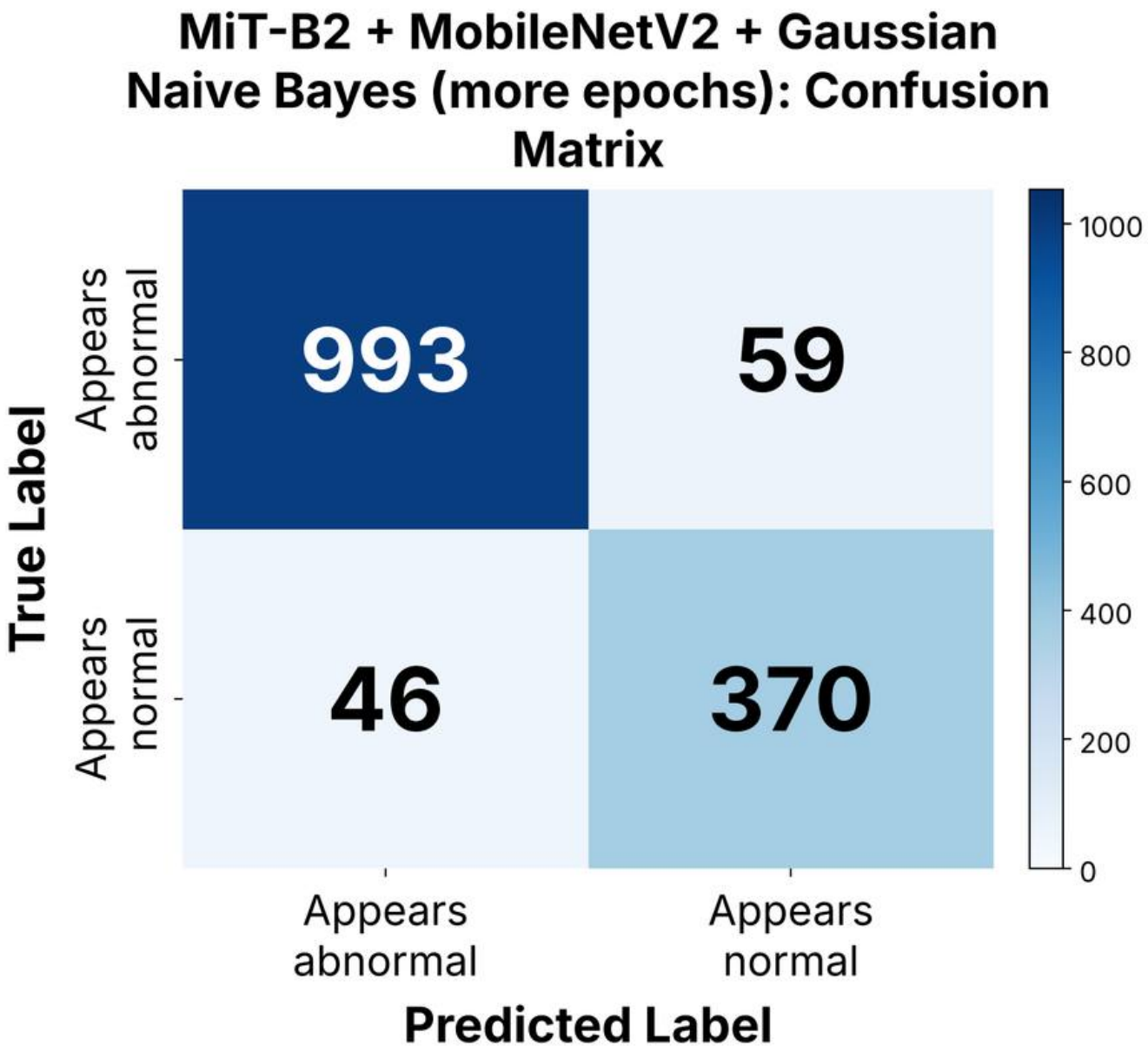
Accuracy: 92.85%

Precision: 92.93%

Recall: 92.85%

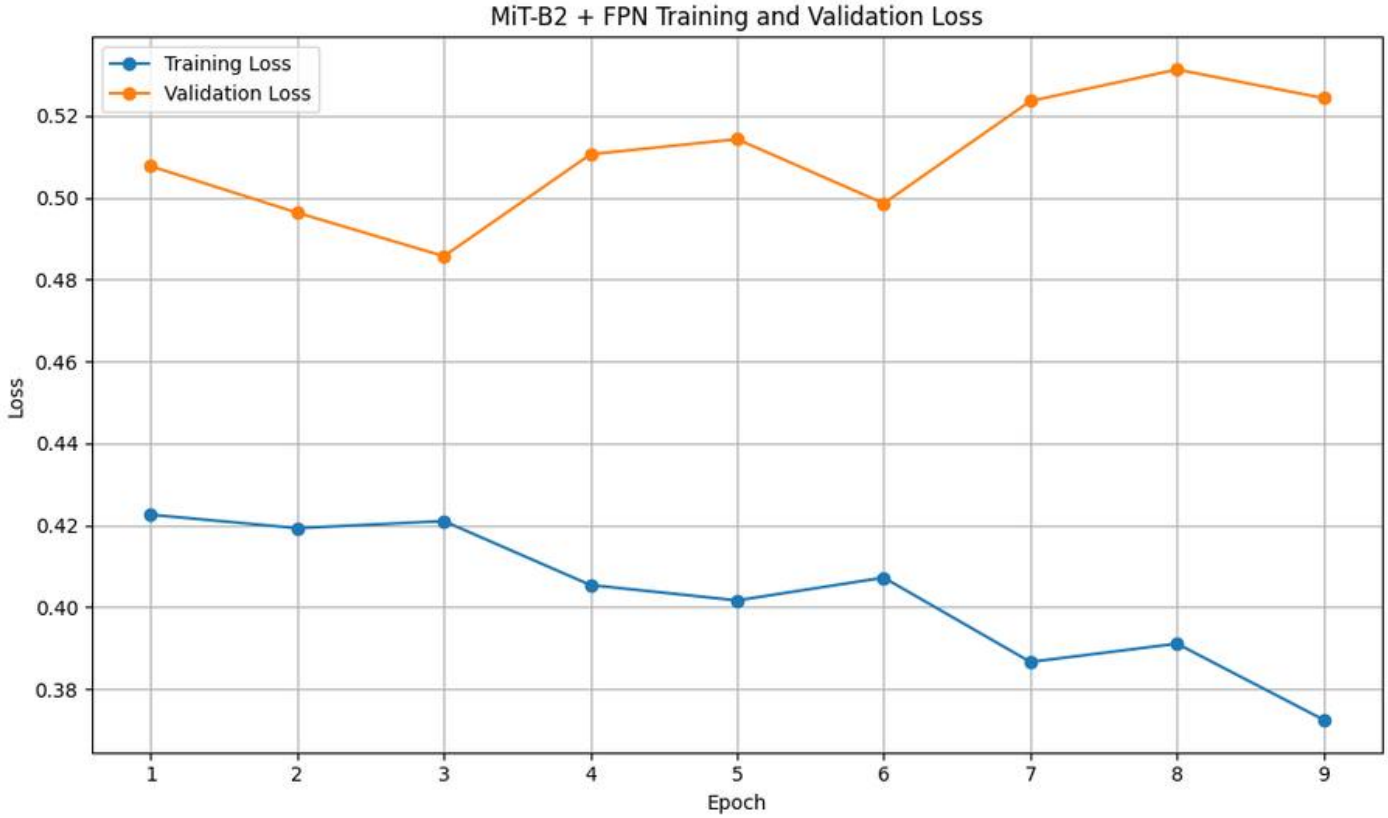
F1-score: 92.88%

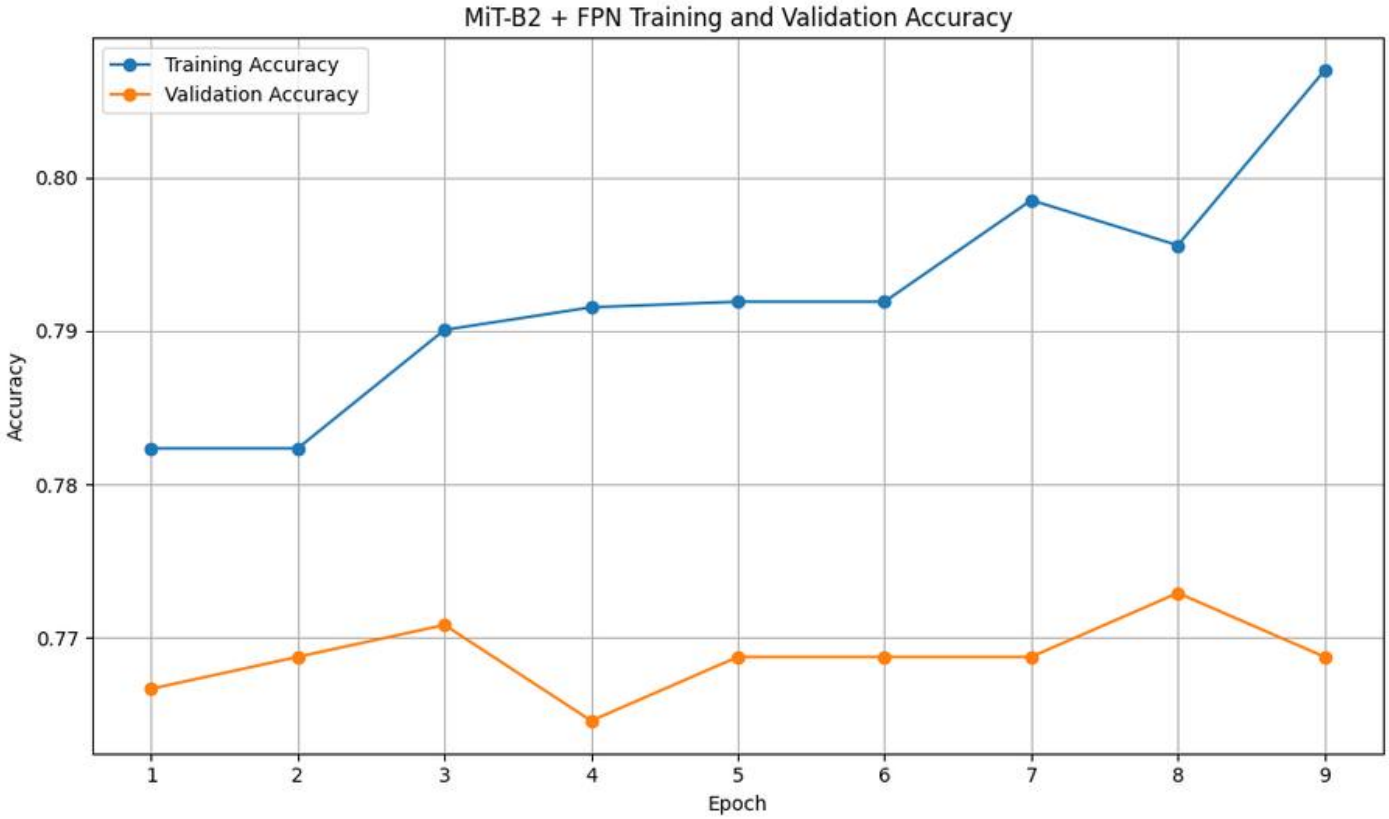
Confusion Matrix



1. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes (more epochs) — graphs (1/2)

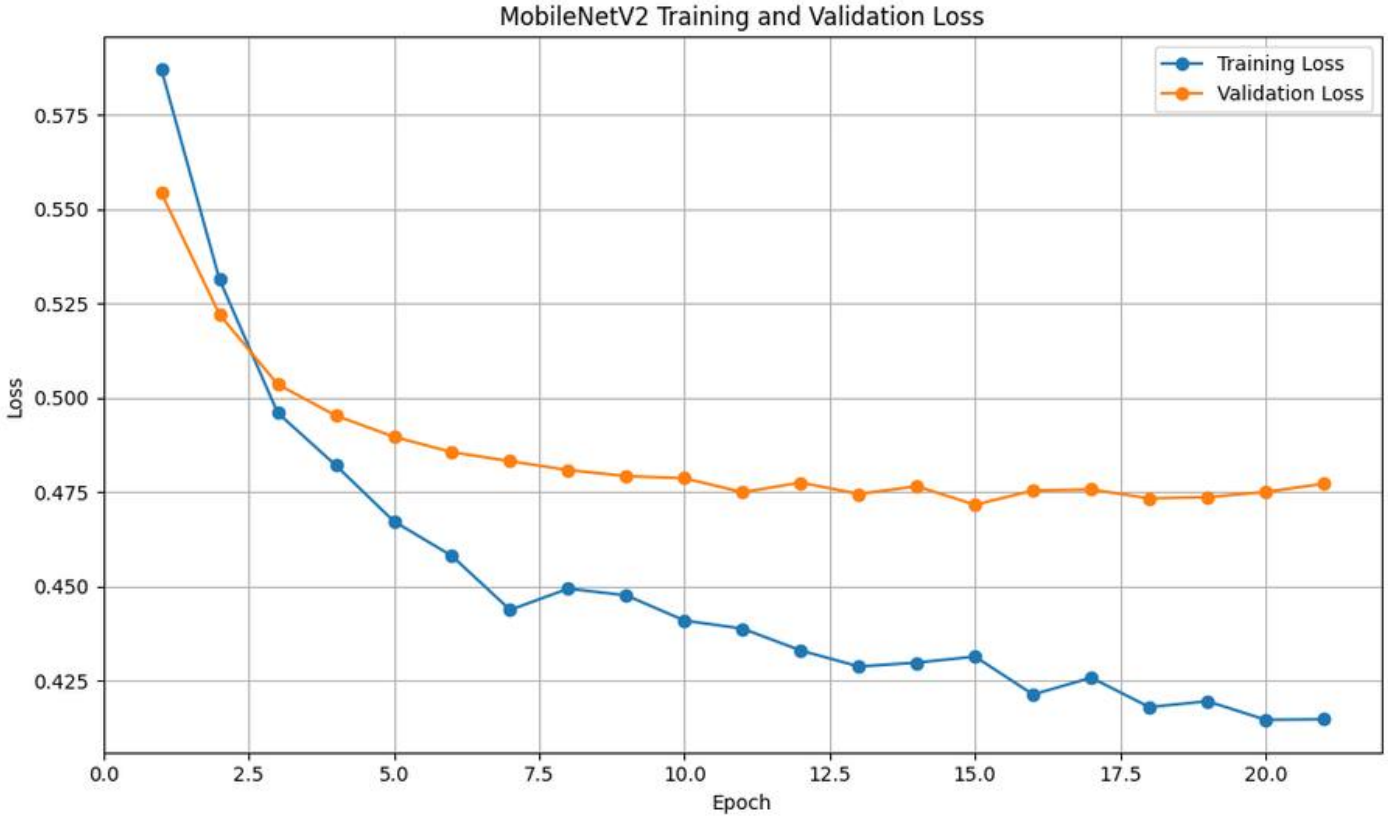
Training / Validation Graphs

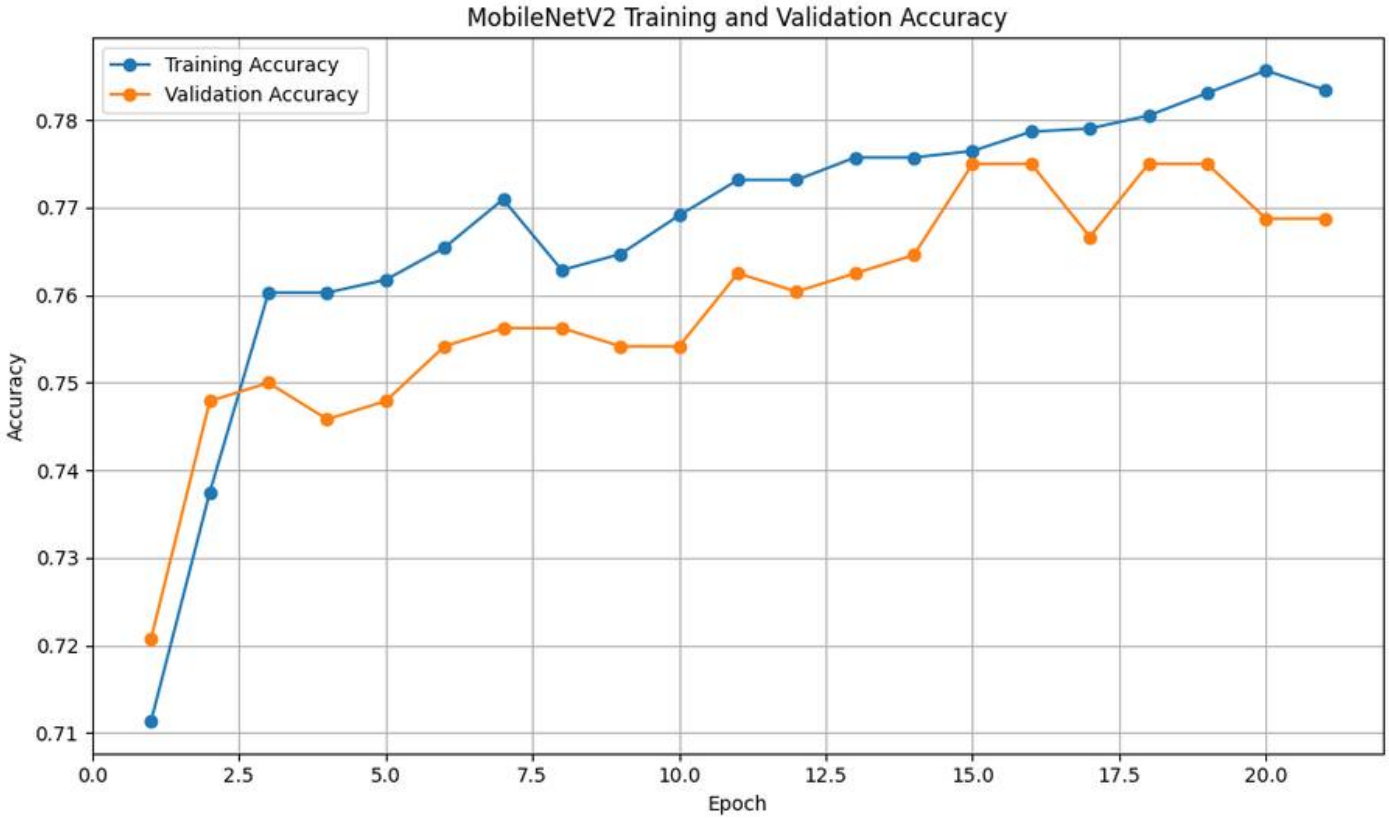




1. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes (more epochs) — graphs (2/2)

Training / Validation Graphs





2. Tuned MiT-B2 + MobileNetV2 + Gaussian Naive Bayes

Source notebook: MIT + MobileNet +NB.ipynb (inside MIT + MobileNet.zip)

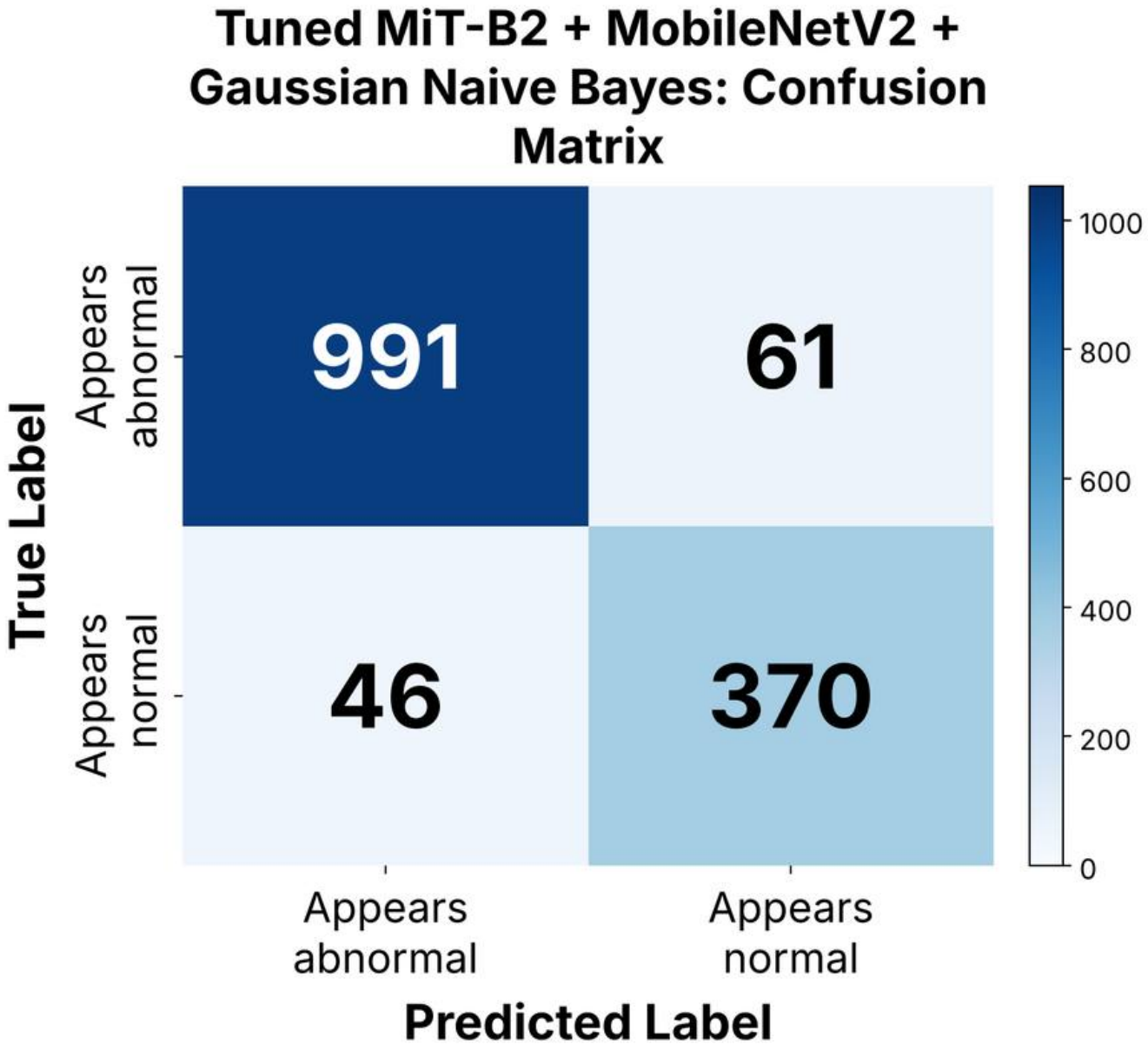
Accuracy: 92.71%

Precision: 92.81%

Recall: 92.71%

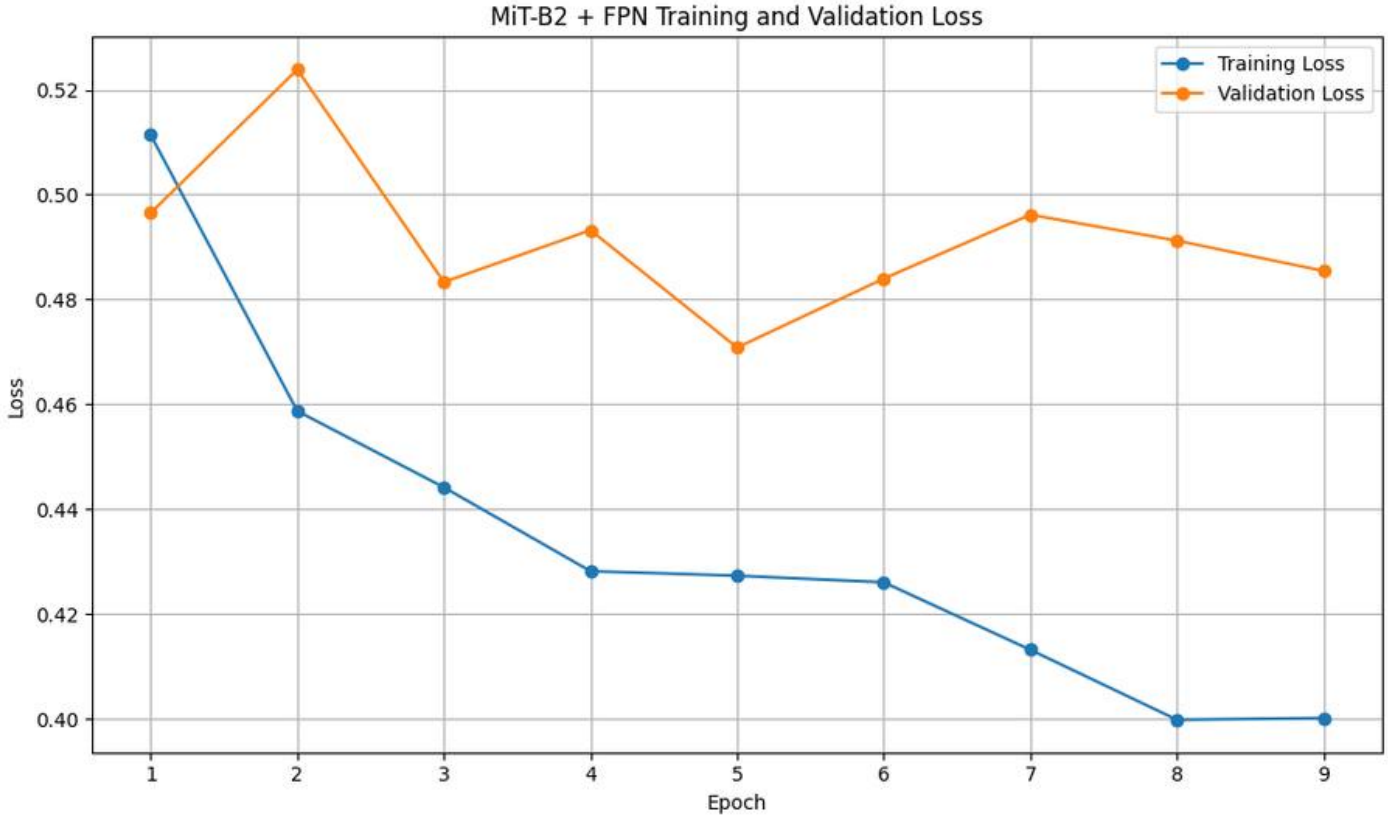
F1-score: 92.75%

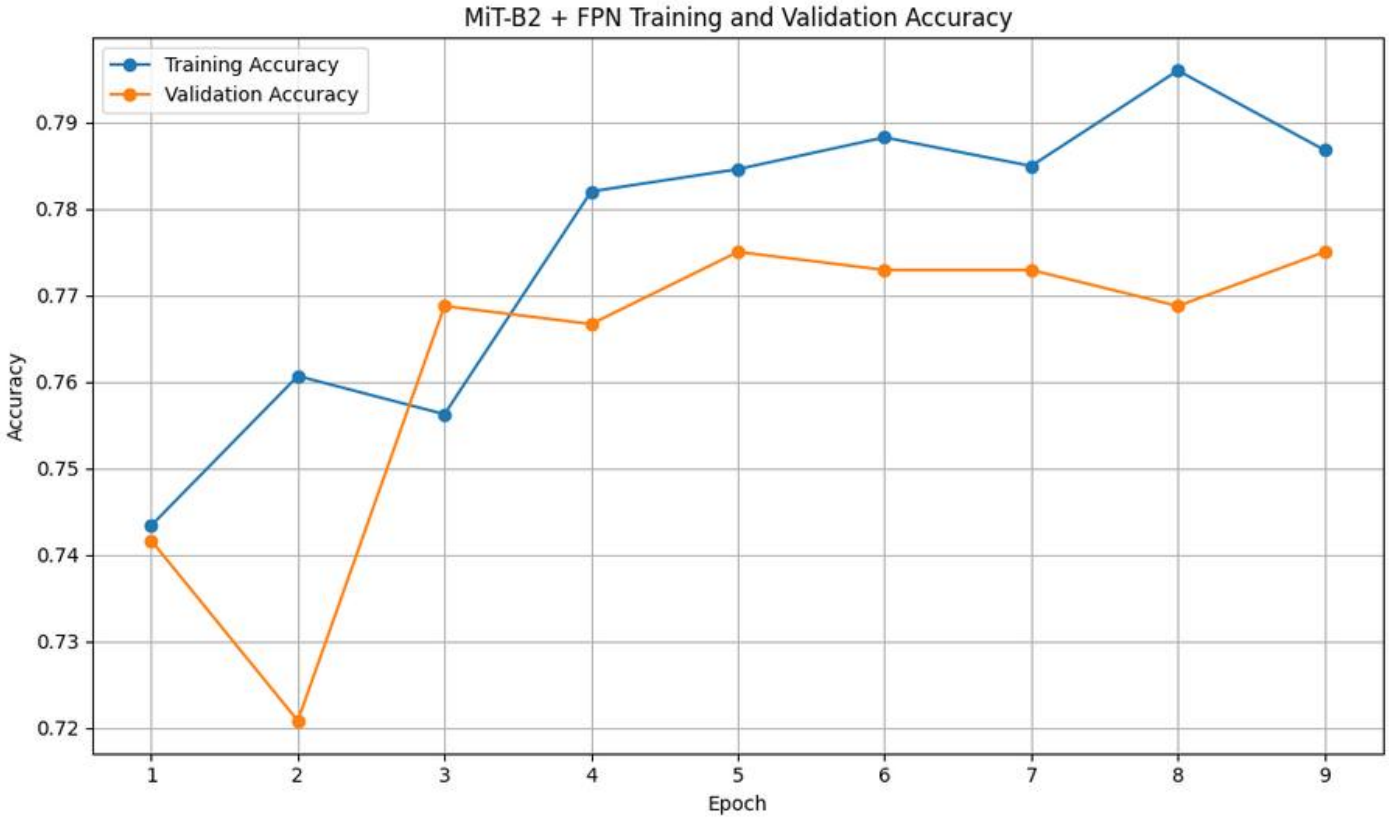
Confusion Matrix



2. Tuned MiT-B2 + MobileNetV2 + Gaussian Naive Bayes — graphs (1/2)

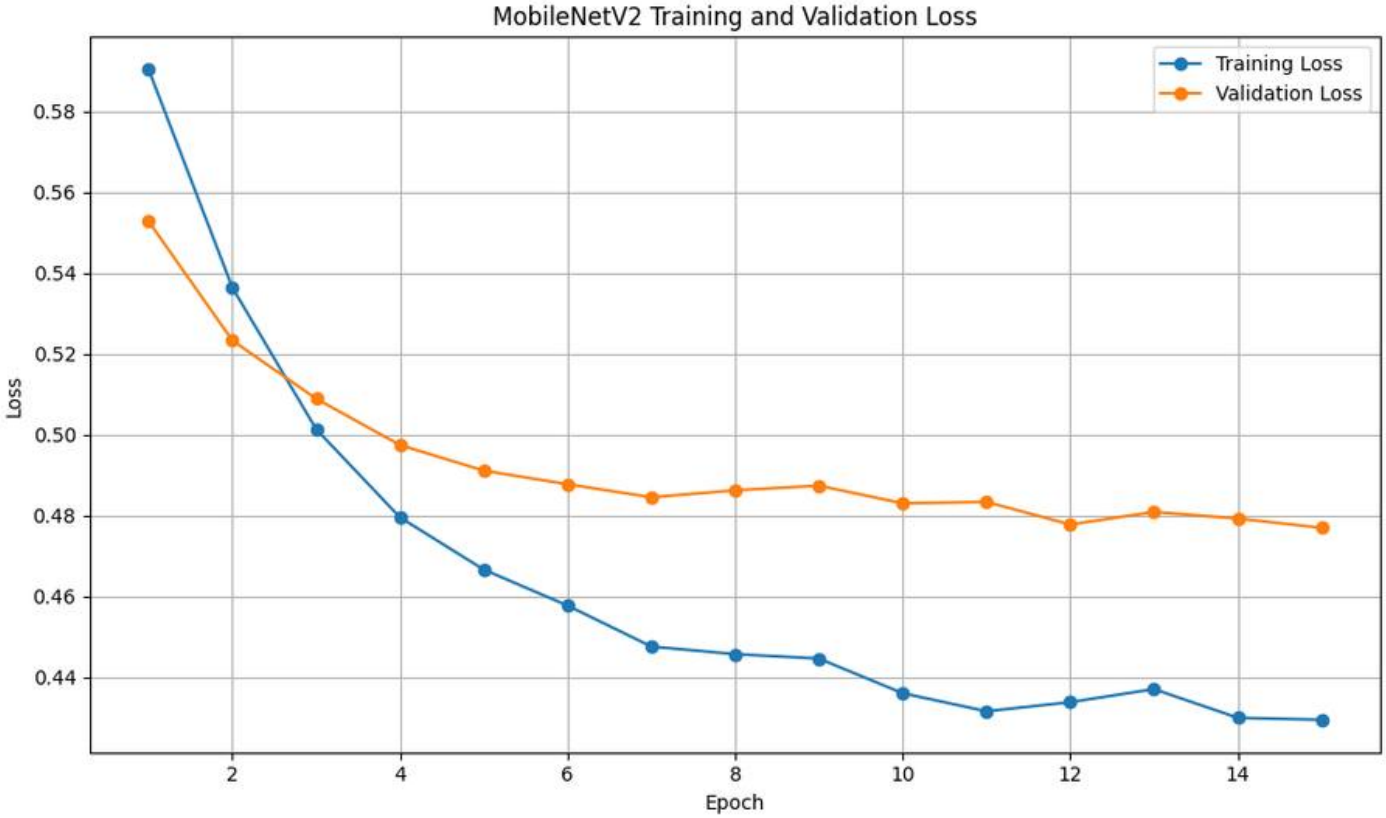
Training / Validation Graphs

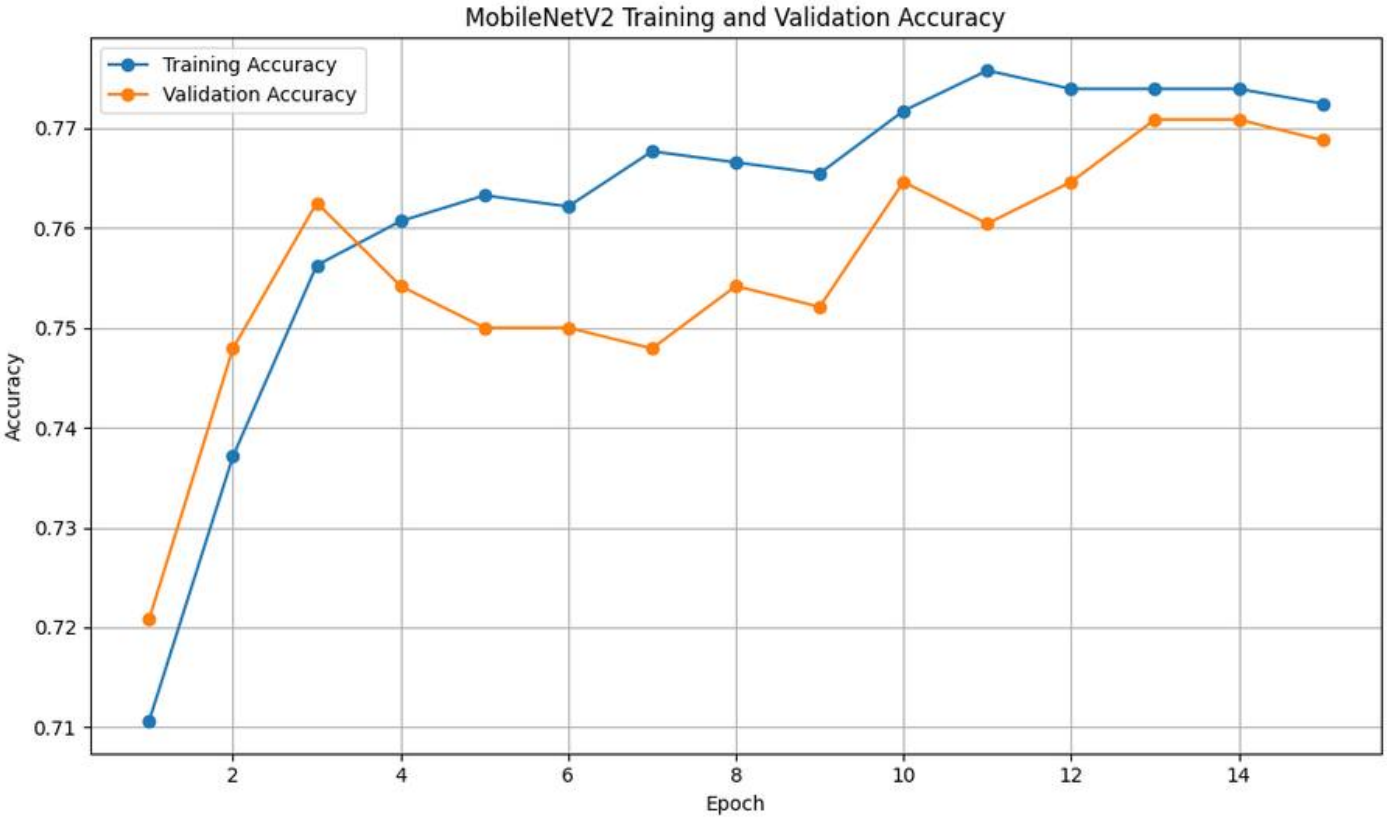




2. Tuned MiT-B2 + MobileNetV2 + Gaussian Naive Bayes — graphs (2/2)

Training / Validation Graphs





3. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes

Source notebook: MIT+ MobileNet+ ML.ipynb (inside MIT + MobileNet.zip)

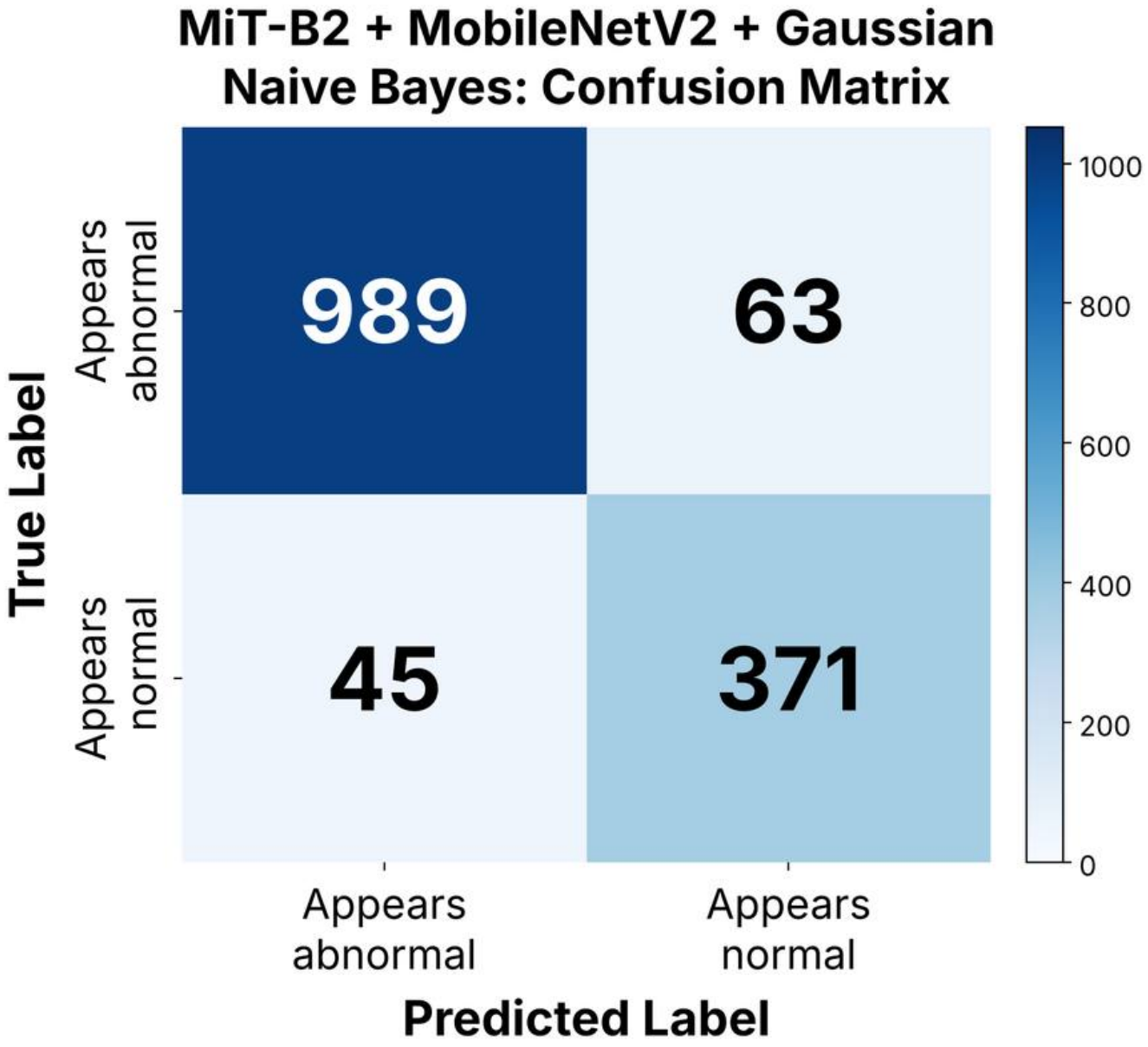
Accuracy: 92.64%

Precision: 92.77%

Recall: 92.64%

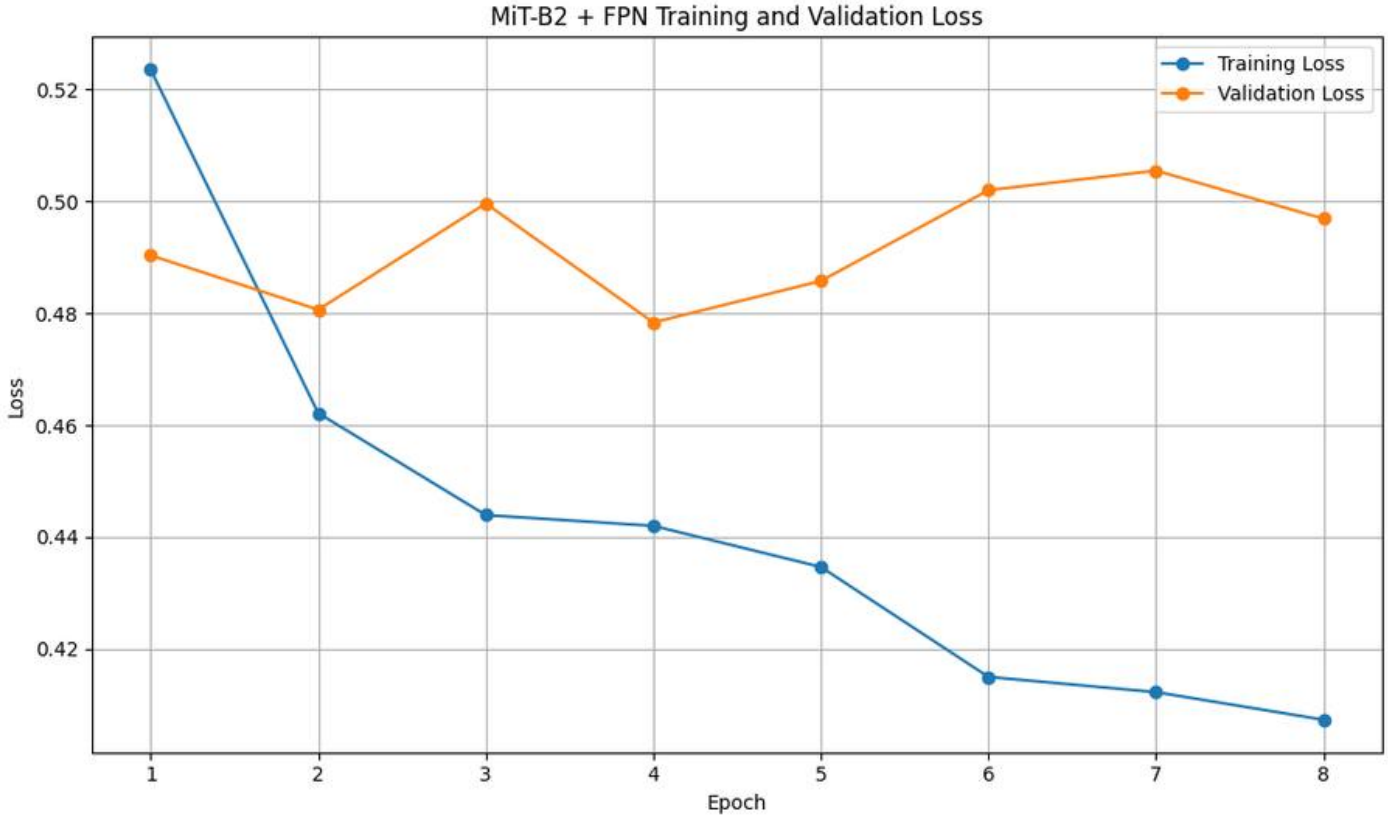
F1-score: 92.69%

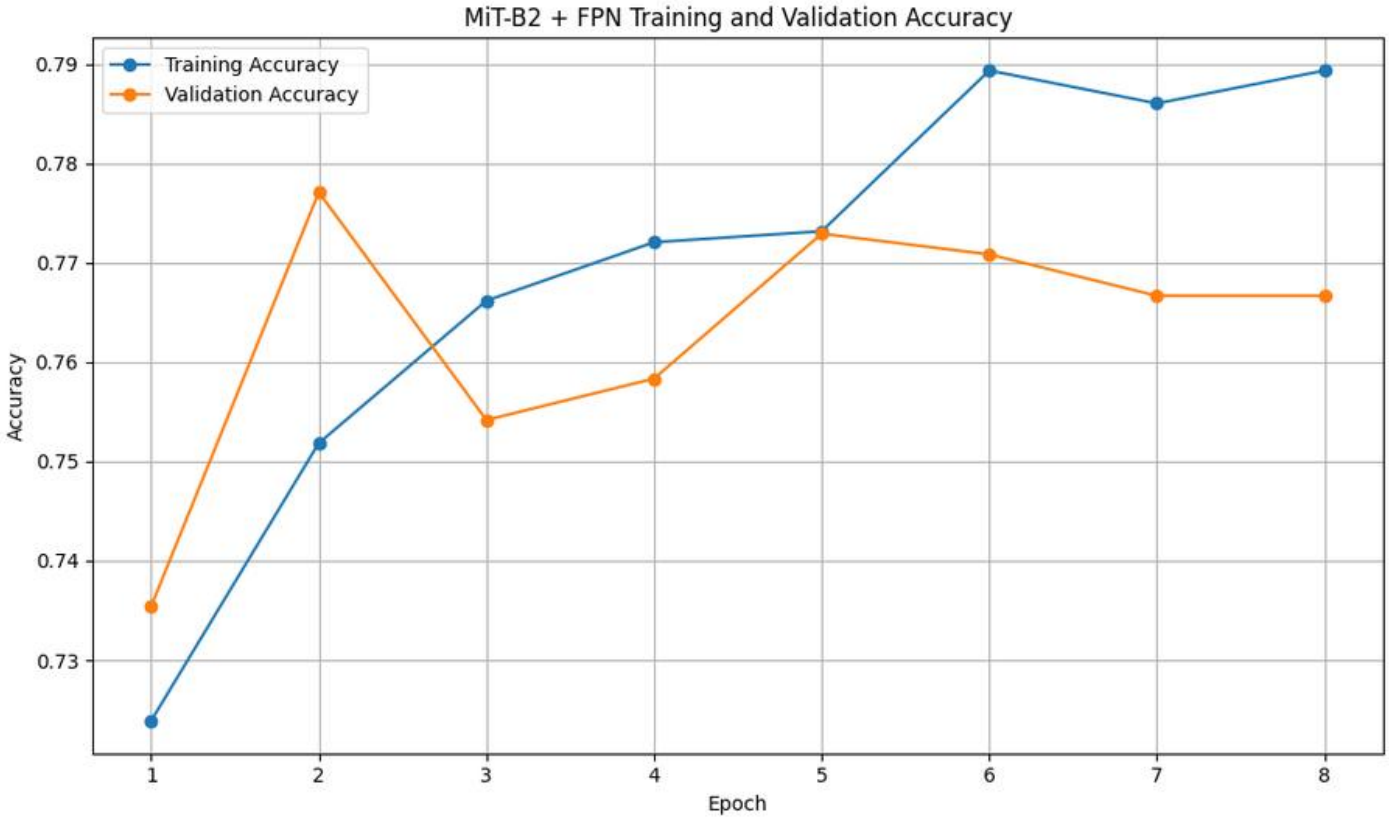
Confusion Matrix



3. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes — graphs (1/2)

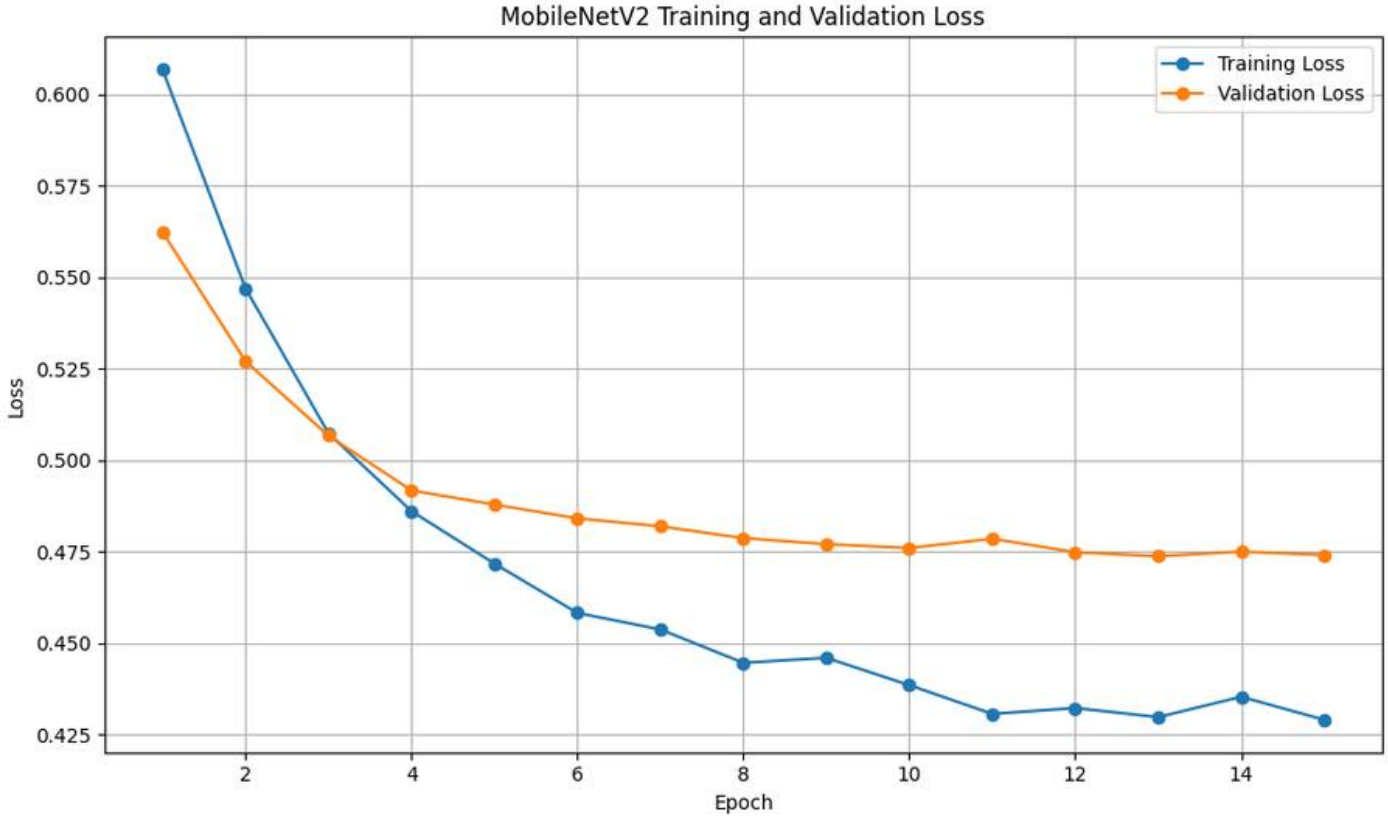
Training / Validation Graphs

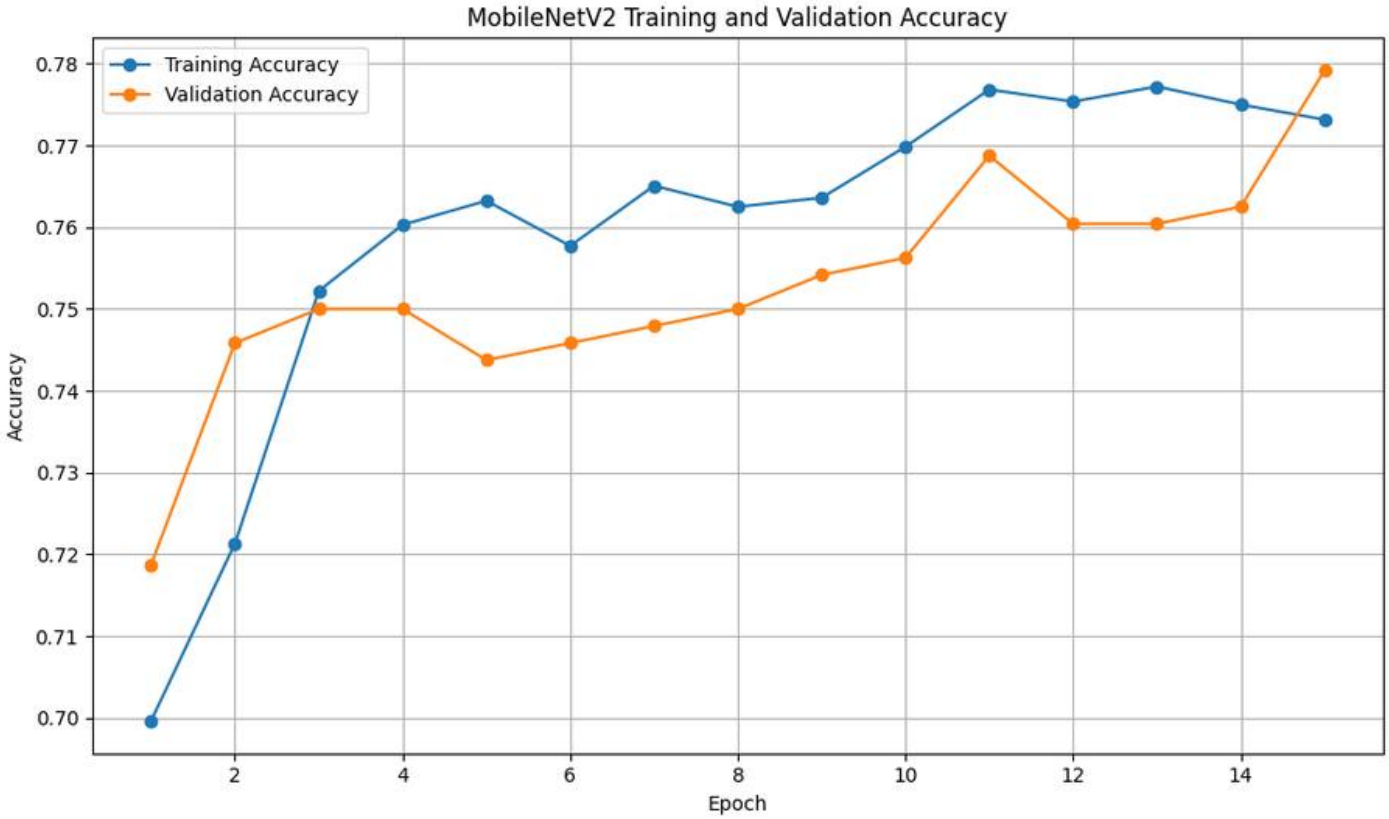




3. MiT-B2 + MobileNetV2 + Gaussian Naive Bayes — graphs (2/2)

Training / Validation Graphs





4. MiT-B2 + MobileNetV2 + KNN (more epochs)

Source notebook: MIT more epochs.ipynb (inside MIT + MobileNet.zip)

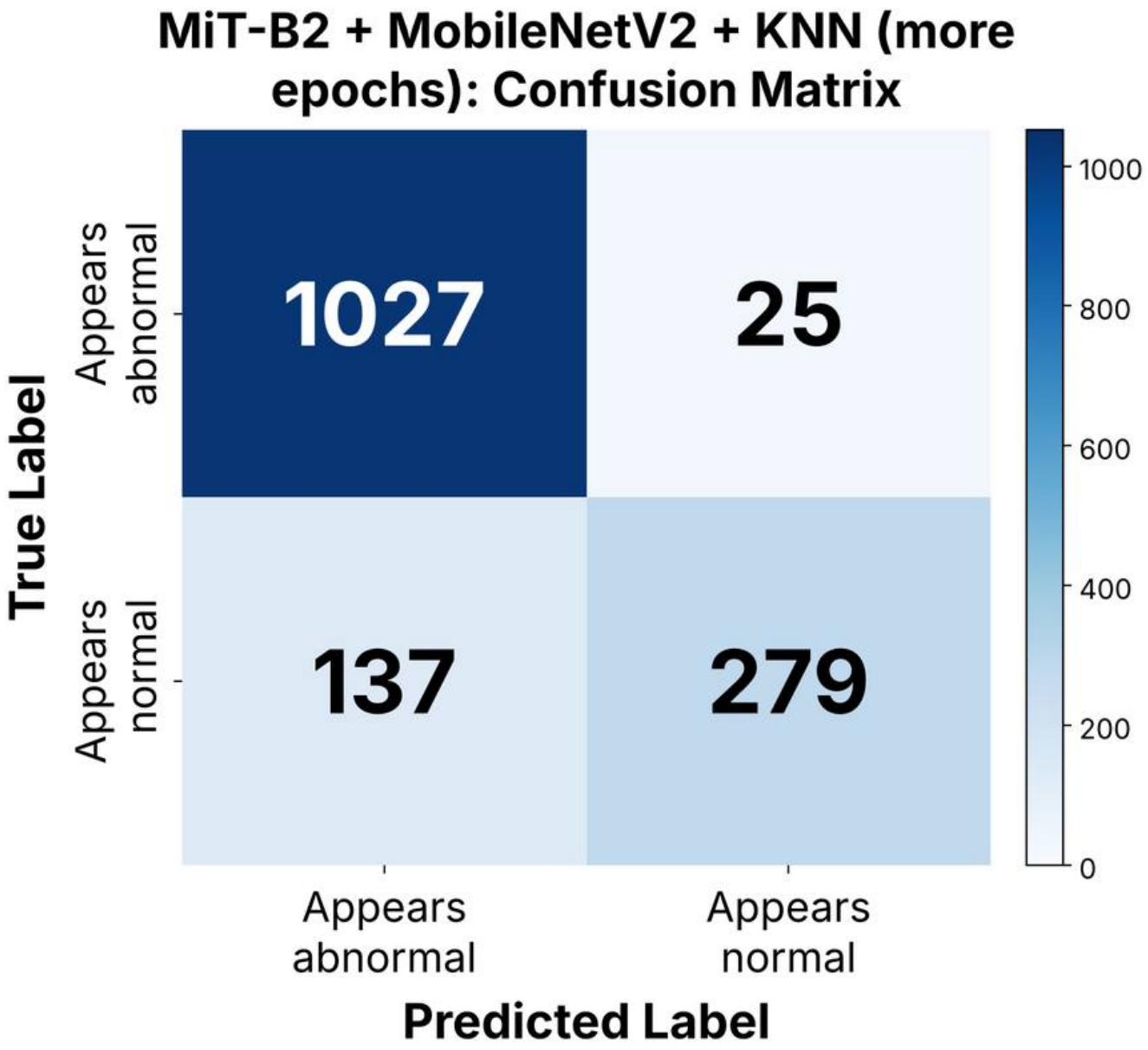
Accuracy: 88.96%

Precision: 89.24%

Recall: 88.96%

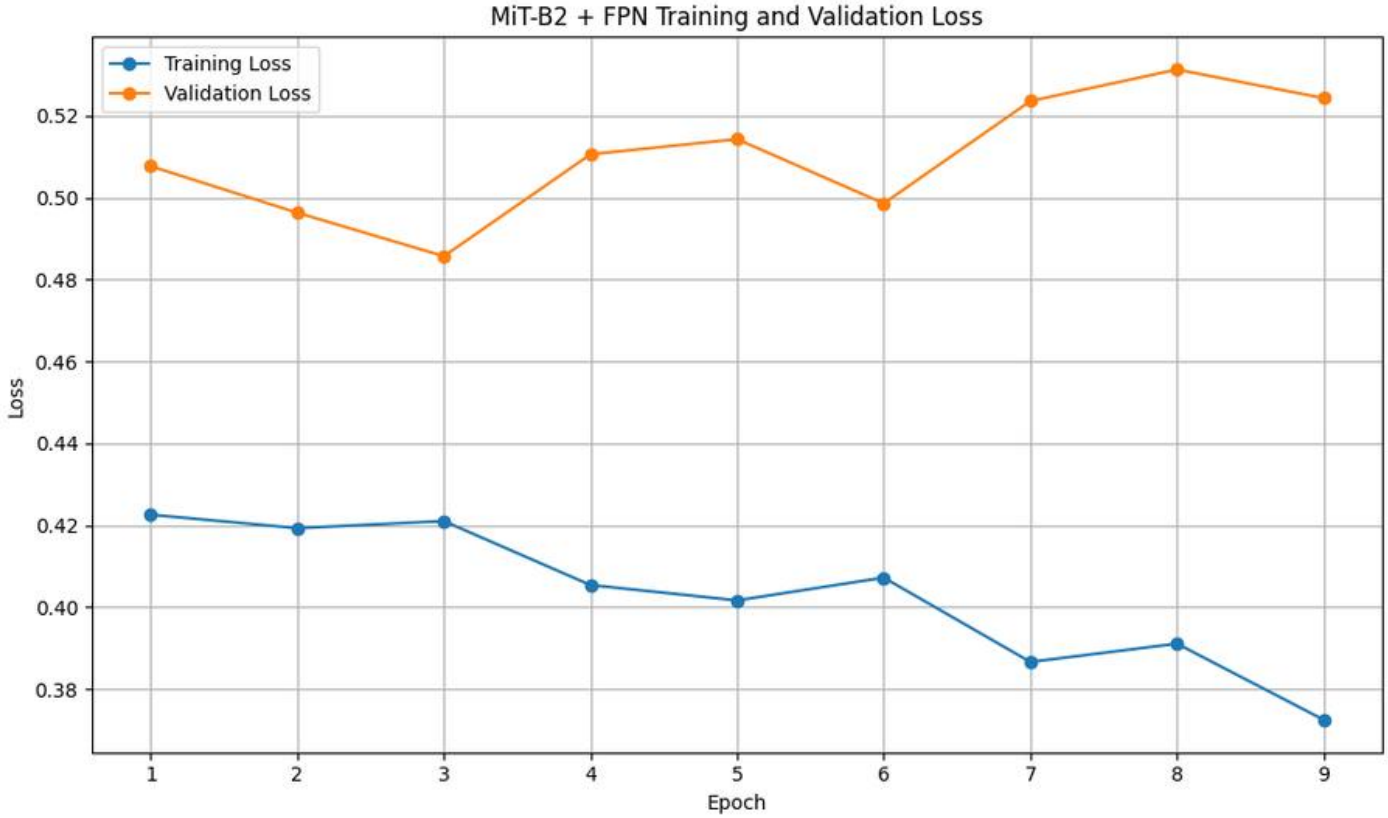
F1-score: 88.39%

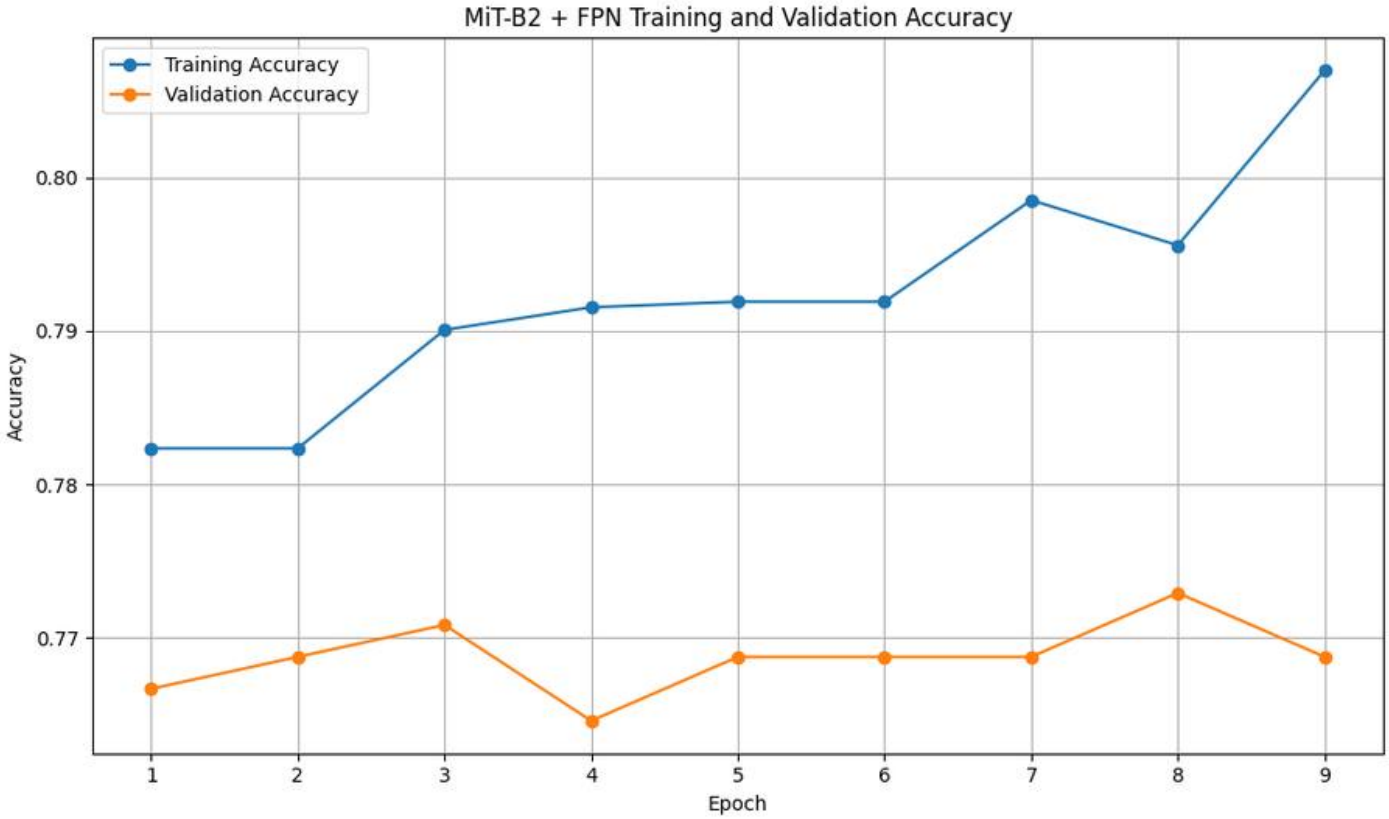
Confusion Matrix



4. MiT-B2 + MobileNetV2 + KNN (more epochs) — graphs (1/2)

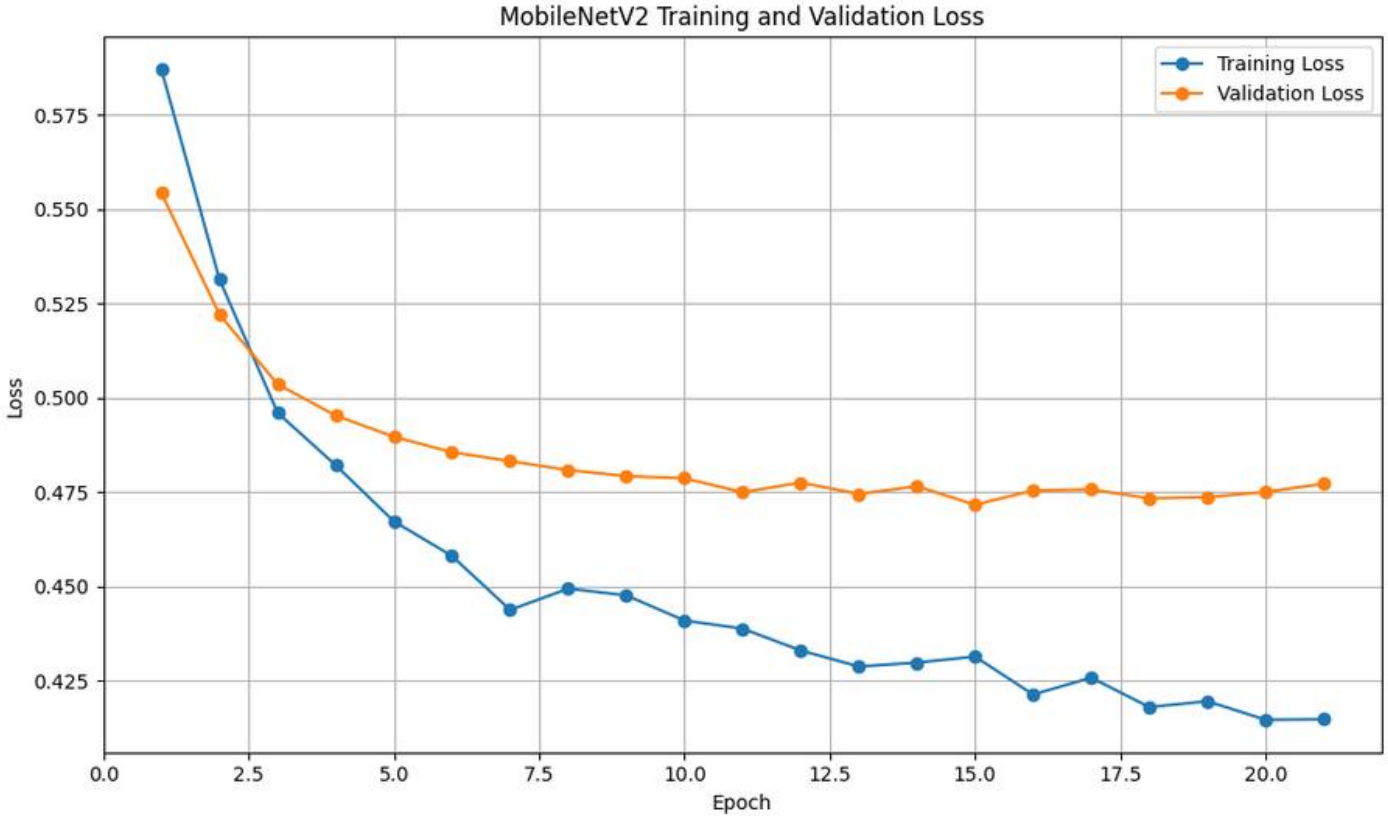
Training / Validation Graphs

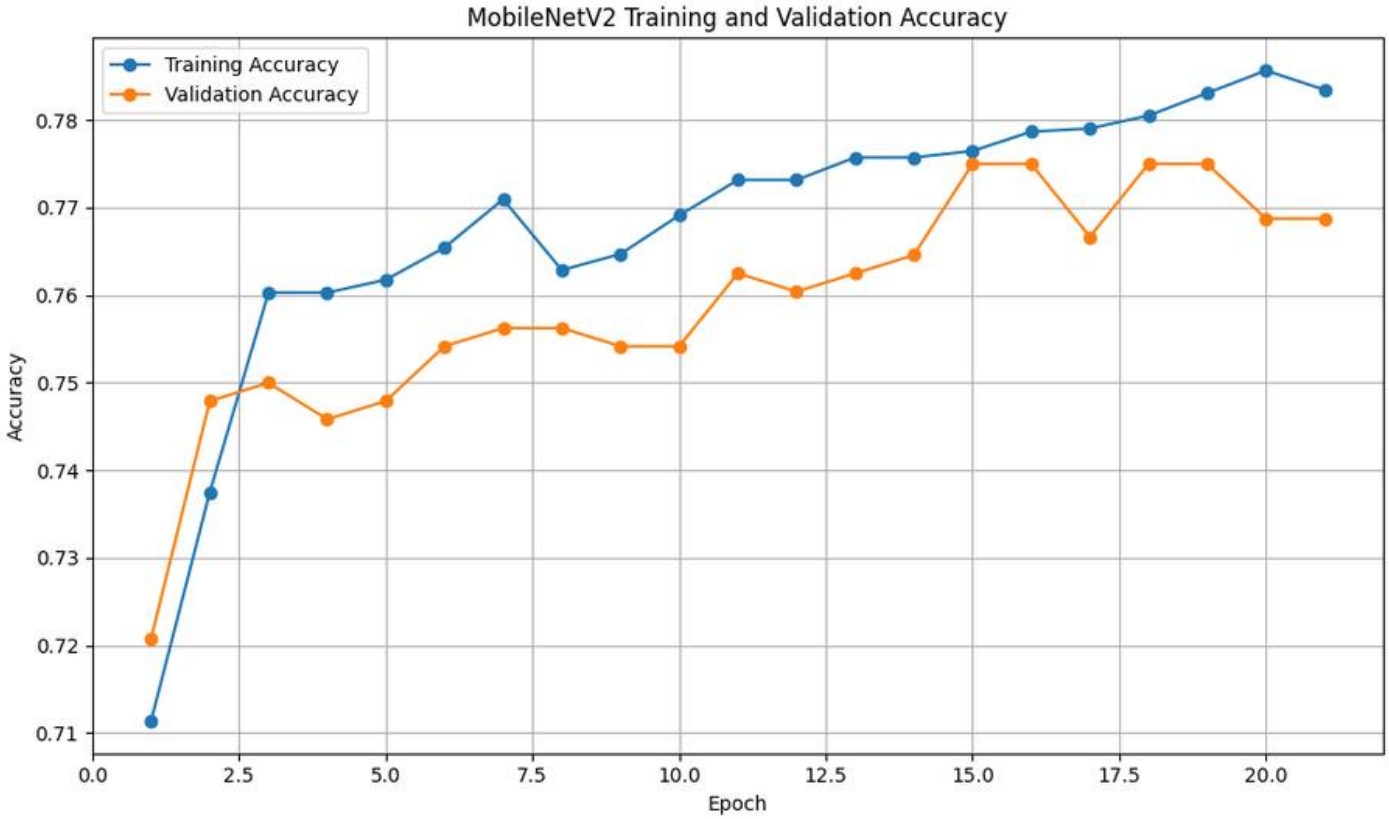




4. MiT-B2 + MobileNetV2 + KNN (more epochs) — graphs (2/2)

Training / Validation Graphs





5. MiT-B2 + MobileNetV2 + KNN

Source notebook: MIT+ MobileNet+ ML.ipynb (inside MIT + MobileNet.zip)

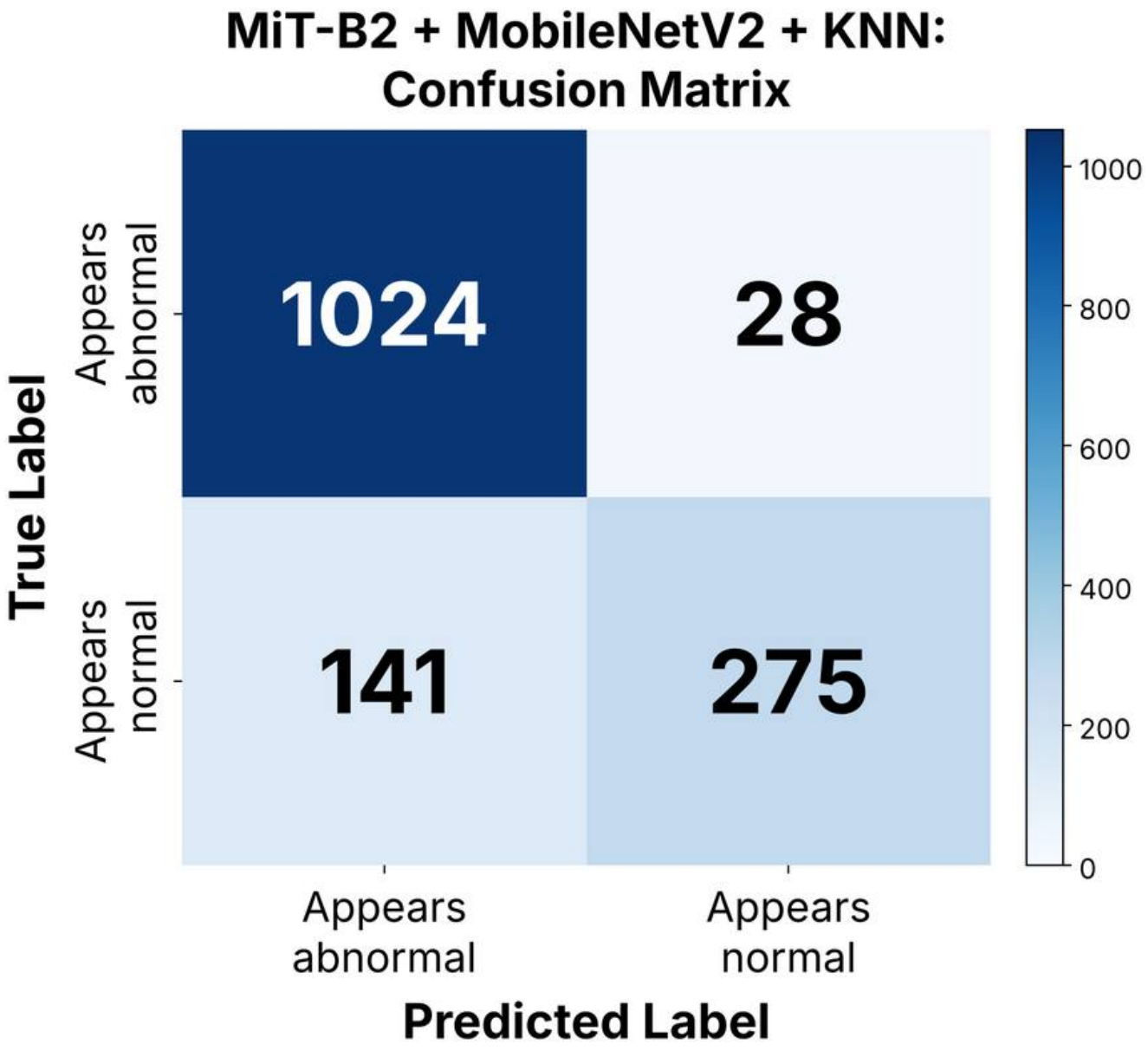
Accuracy: 88.49%

Precision: 88.71%

Recall: 88.49%

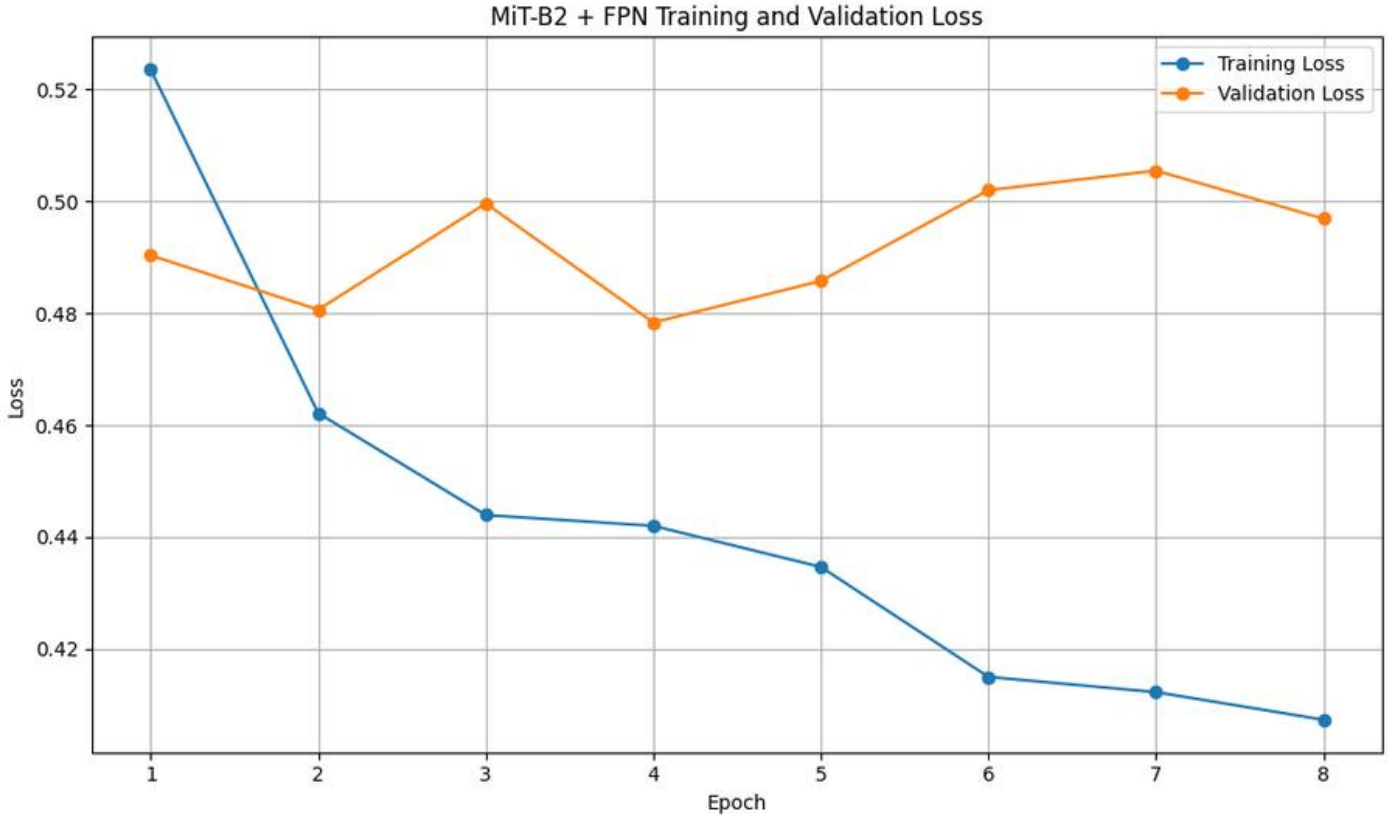
F1-score: 87.88%

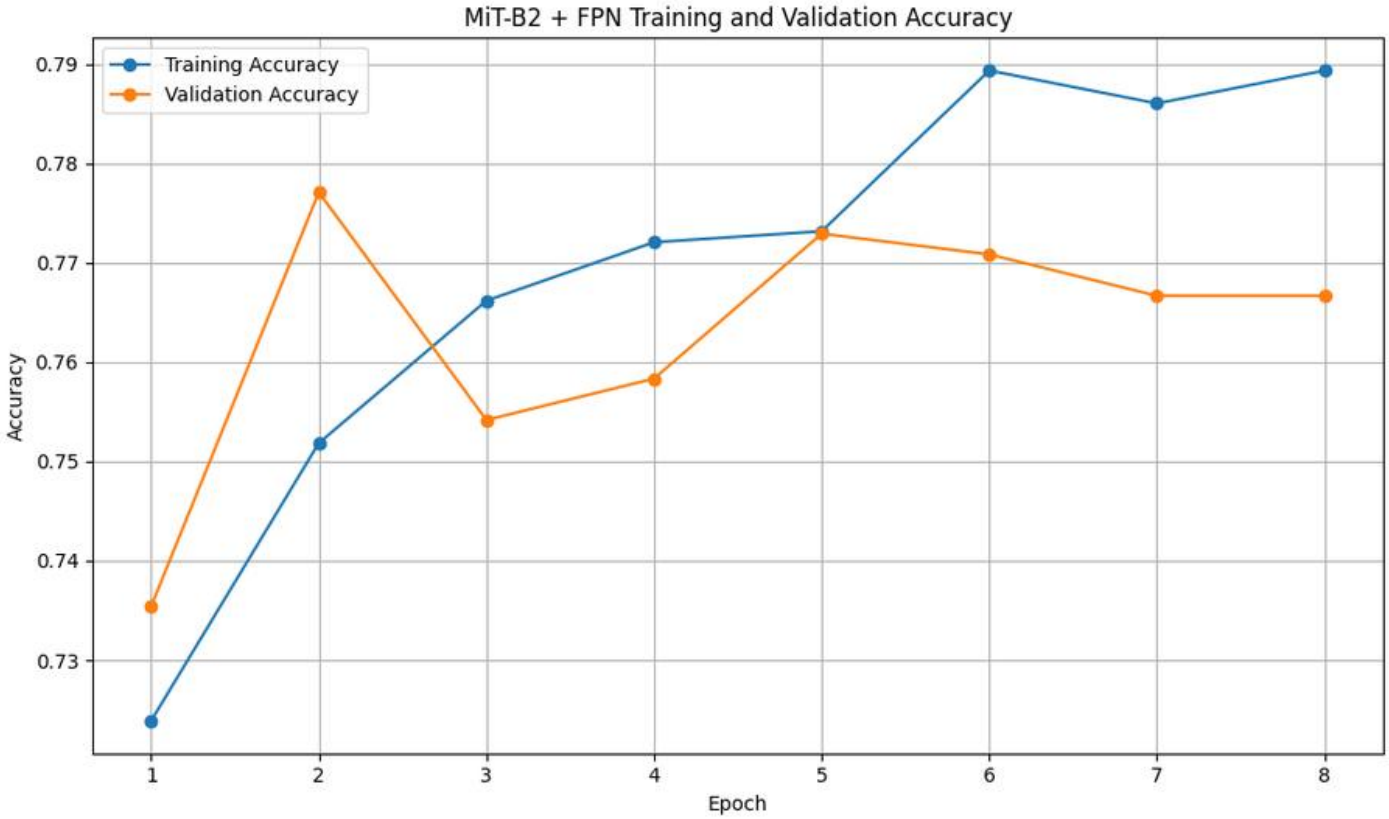
Confusion Matrix



5. MiT-B2 + MobileNetV2 + KNN — graphs (1/2)

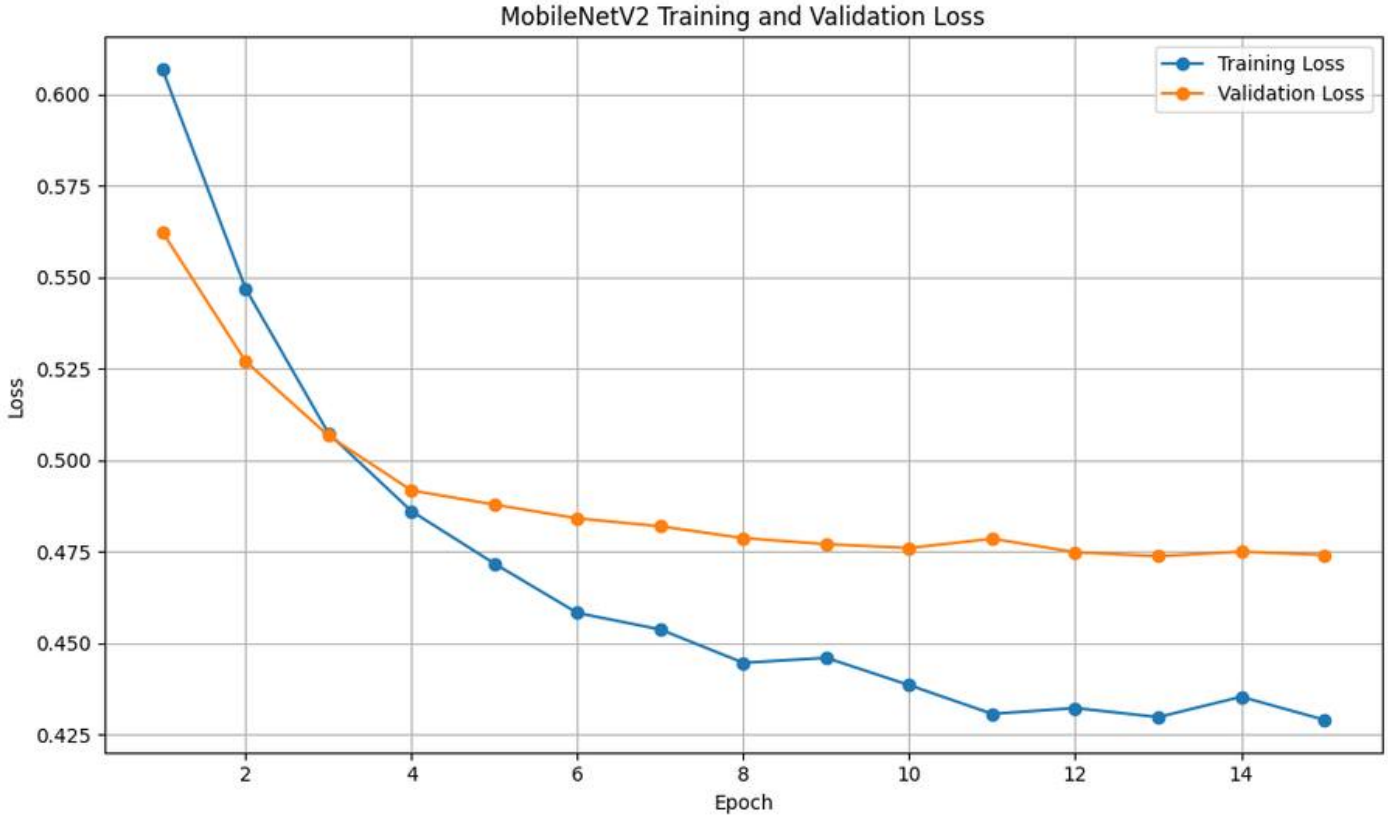
Training / Validation Graphs

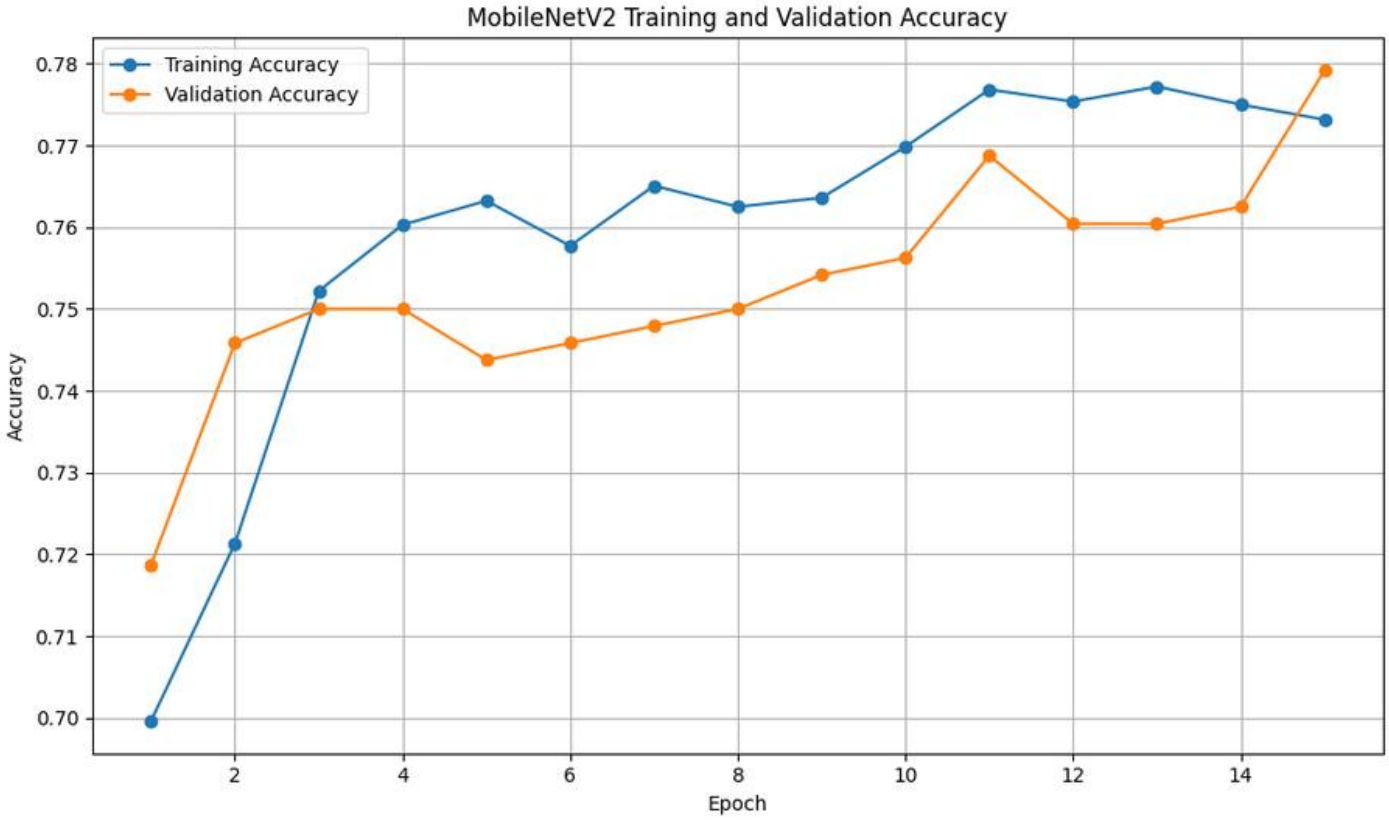




5. MiT-B2 + MobileNetV2 + KNN — graphs (2/2)

Training / Validation Graphs





6. MiT-B2 + FPN + Random Forest

Source notebook: MIT + RF and MIT only.ipynb (inside MIT + MobileNet.zip)

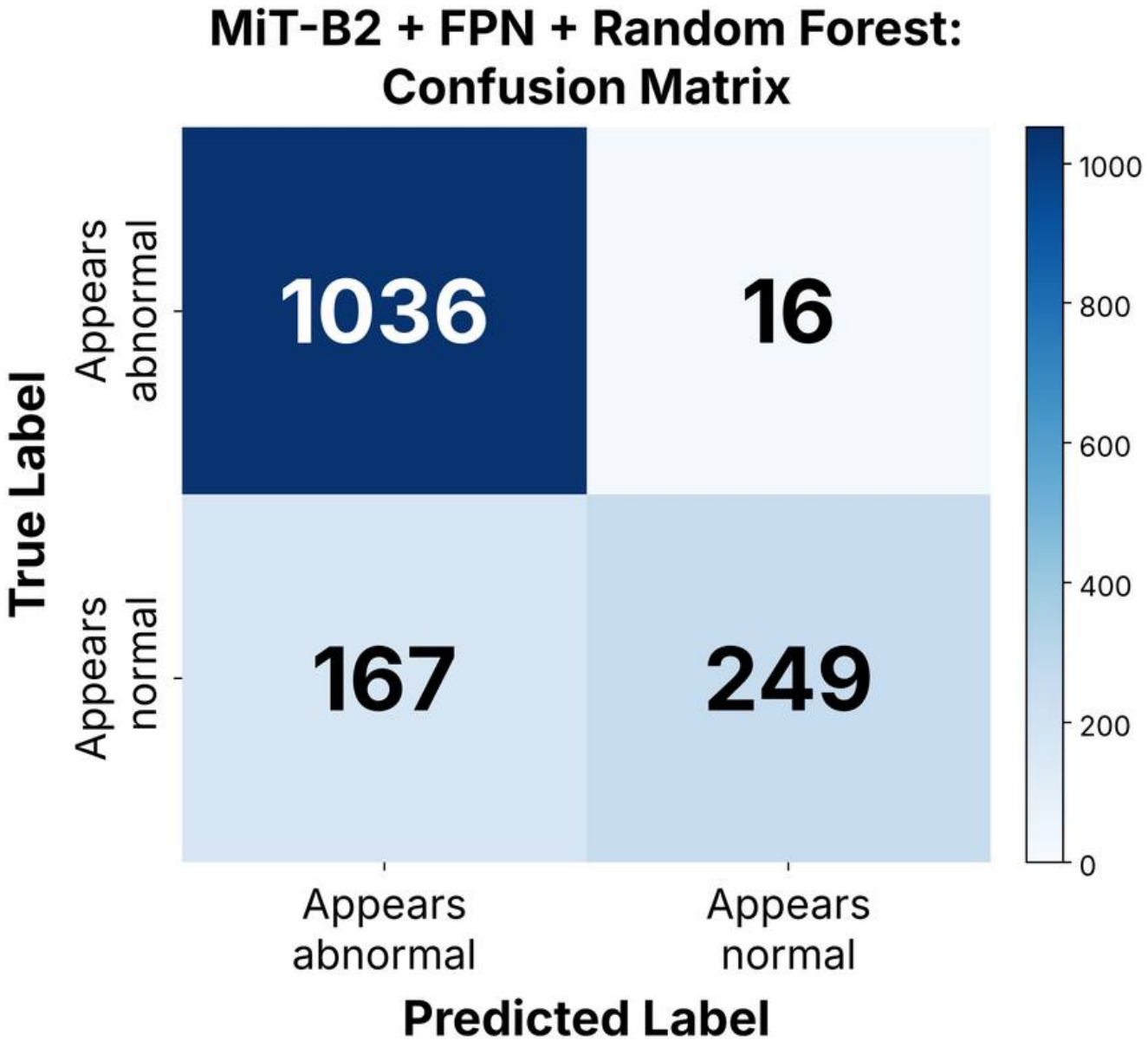
Accuracy: 87.53%

Precision: 88.34%

Recall: 87.53%

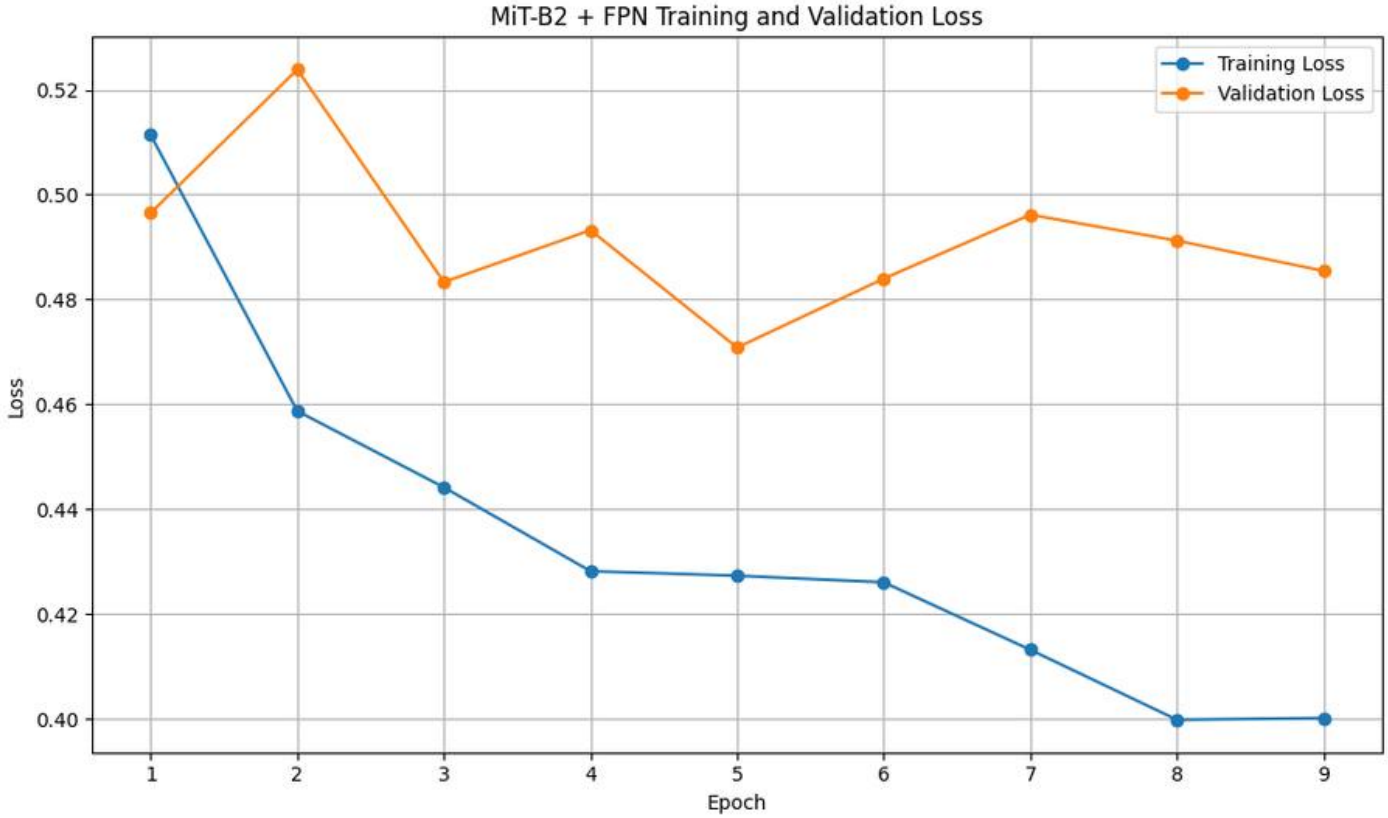
F1-score: 86.57%

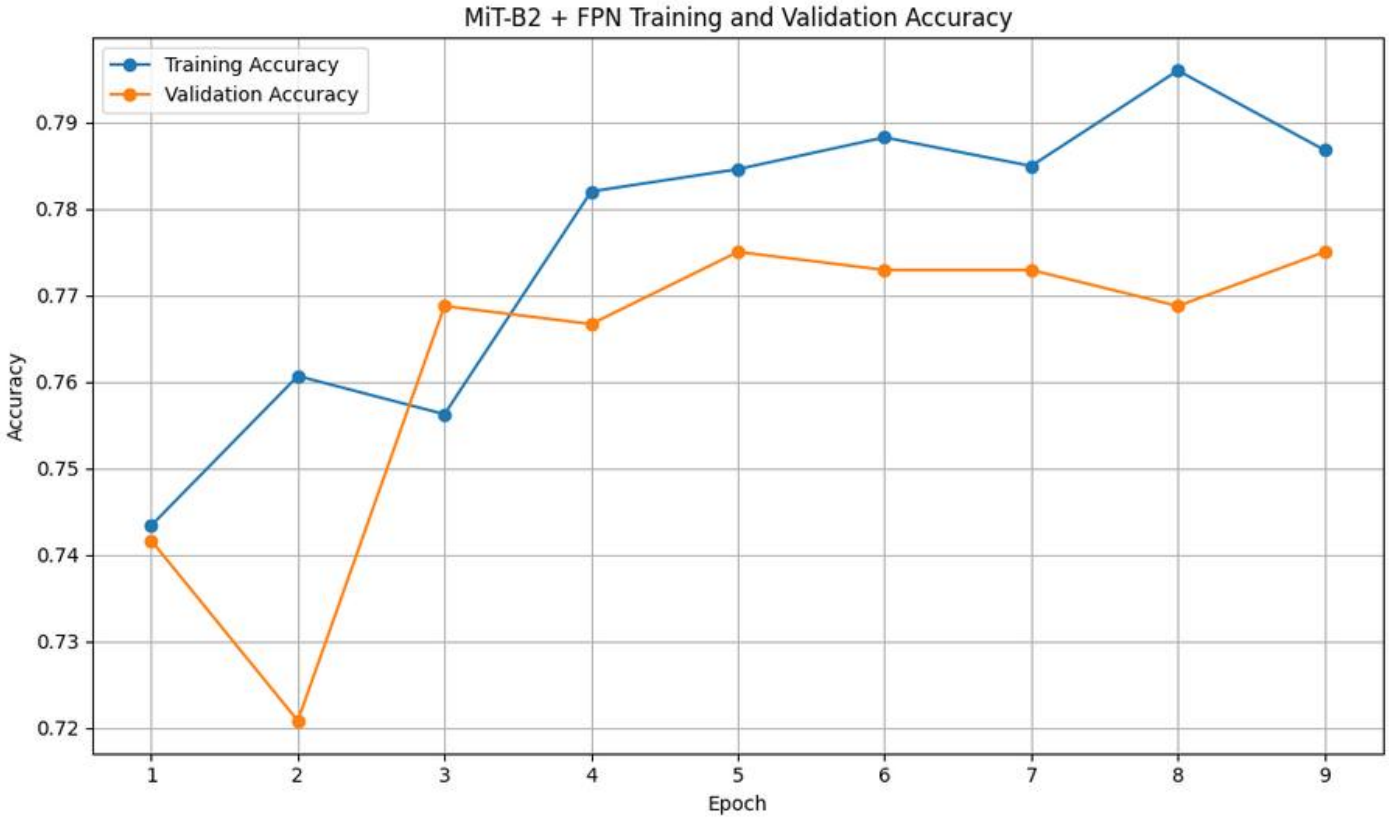
Confusion Matrix



6. MiT-B2 + FPN + Random Forest — graphs

Training / Validation Graphs





7. MiT-B2 + MobileNetV2 + Random Forest

Source notebook: MIT+ MobileNet+ ML.ipynb (inside MIT + MobileNet.zip)

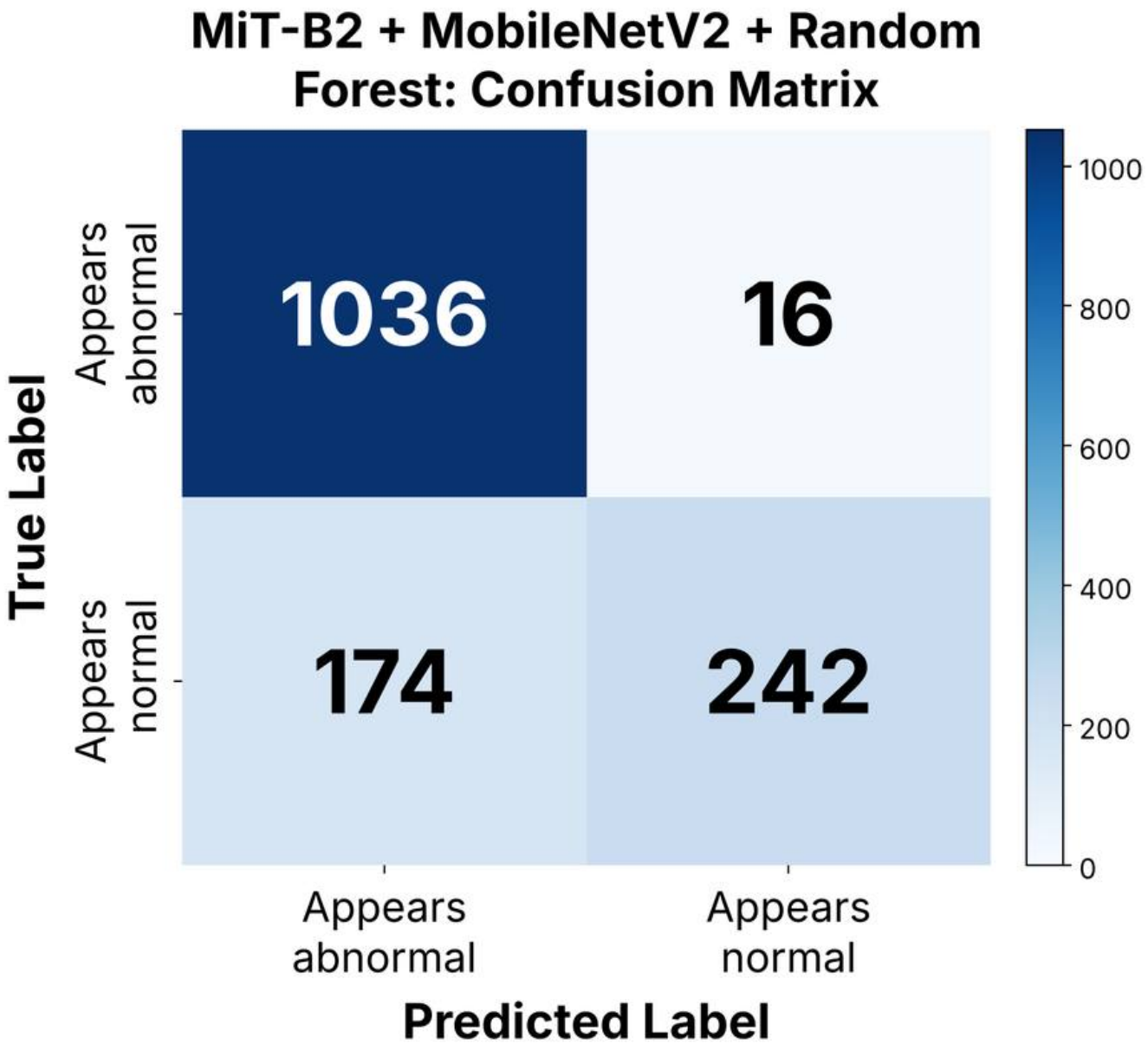
Accuracy: 87.06%

Precision: 87.94%

Recall: 87.06%

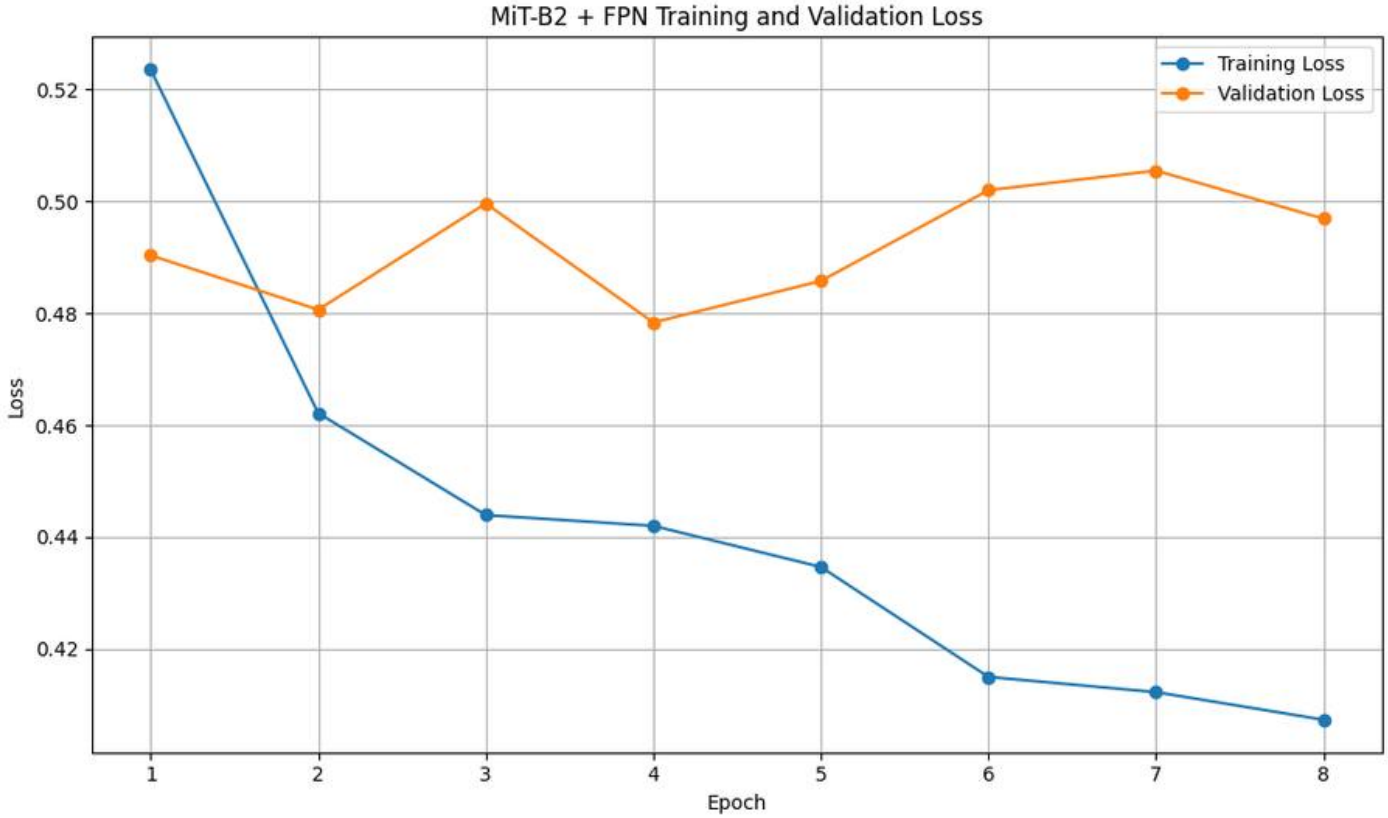
F1-score: 85.99%

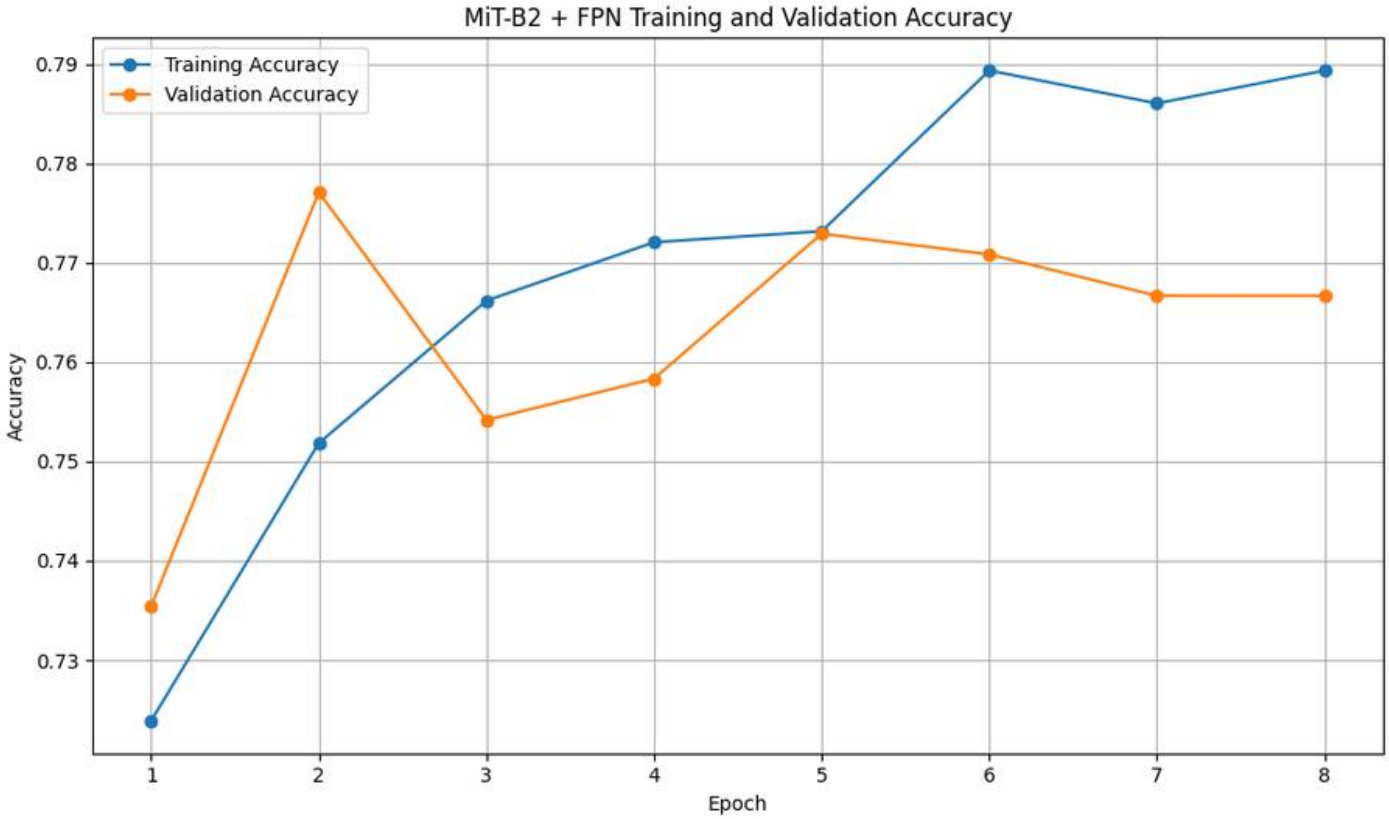
Confusion Matrix



7. MiT-B2 + MobileNetV2 + Random Forest — graphs (1/2)

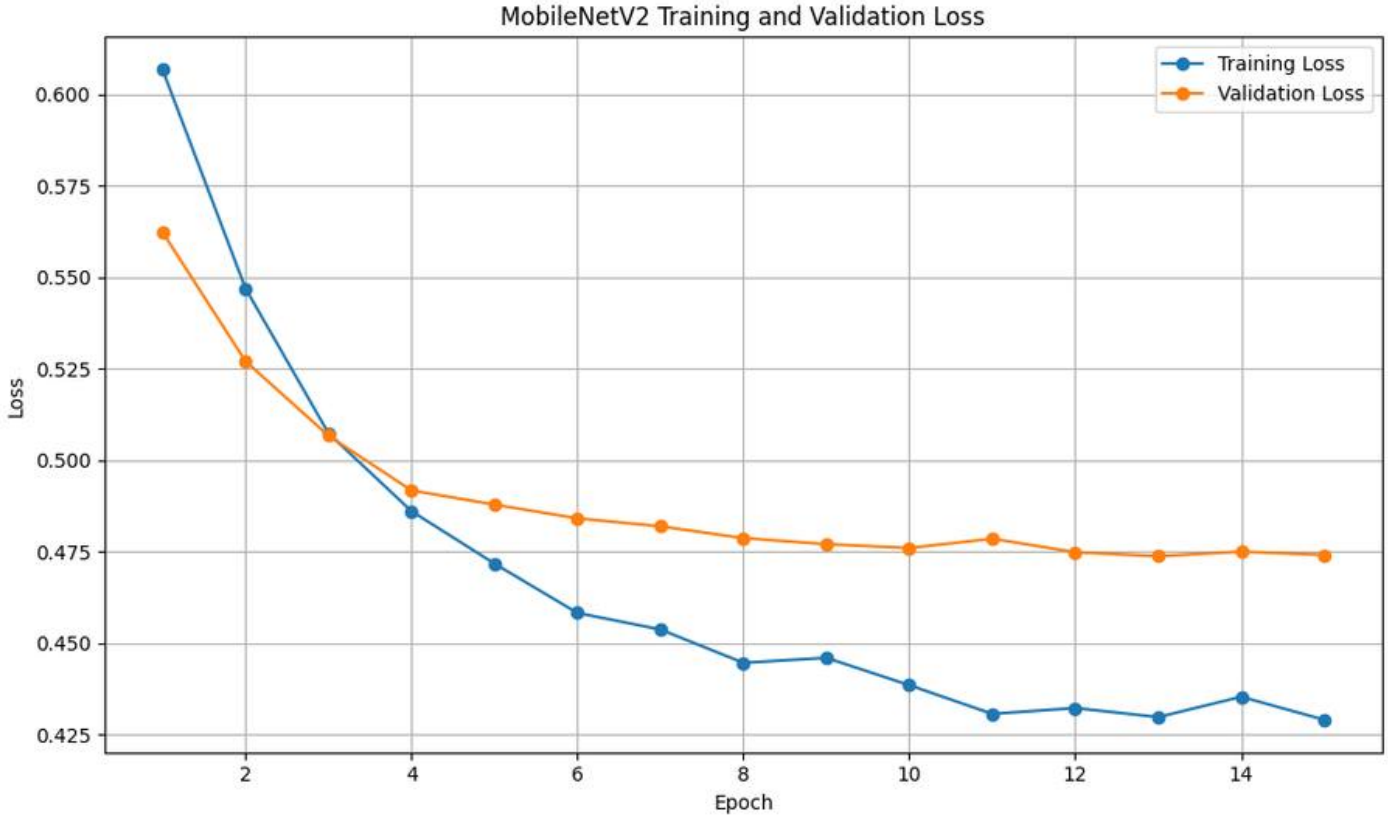
Training / Validation Graphs

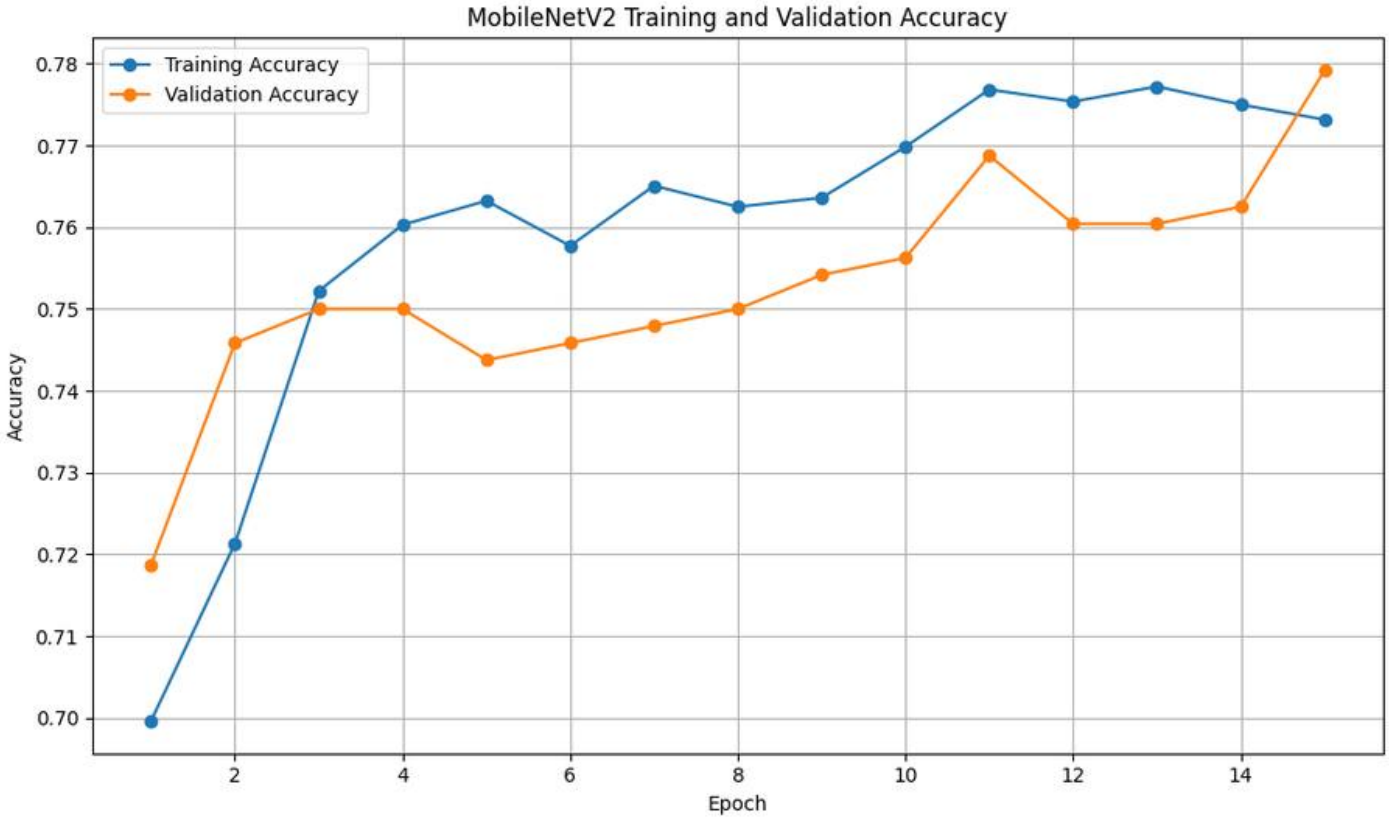




7. MiT-B2 + MobileNetV2 + Random Forest — graphs (2/2)

Training / Validation Graphs





8. MiT-B2 + MobileNetV2 + Random Forest (more epochs)

Source notebook: MIT more epochs.ipynb (inside MIT + MobileNet.zip)

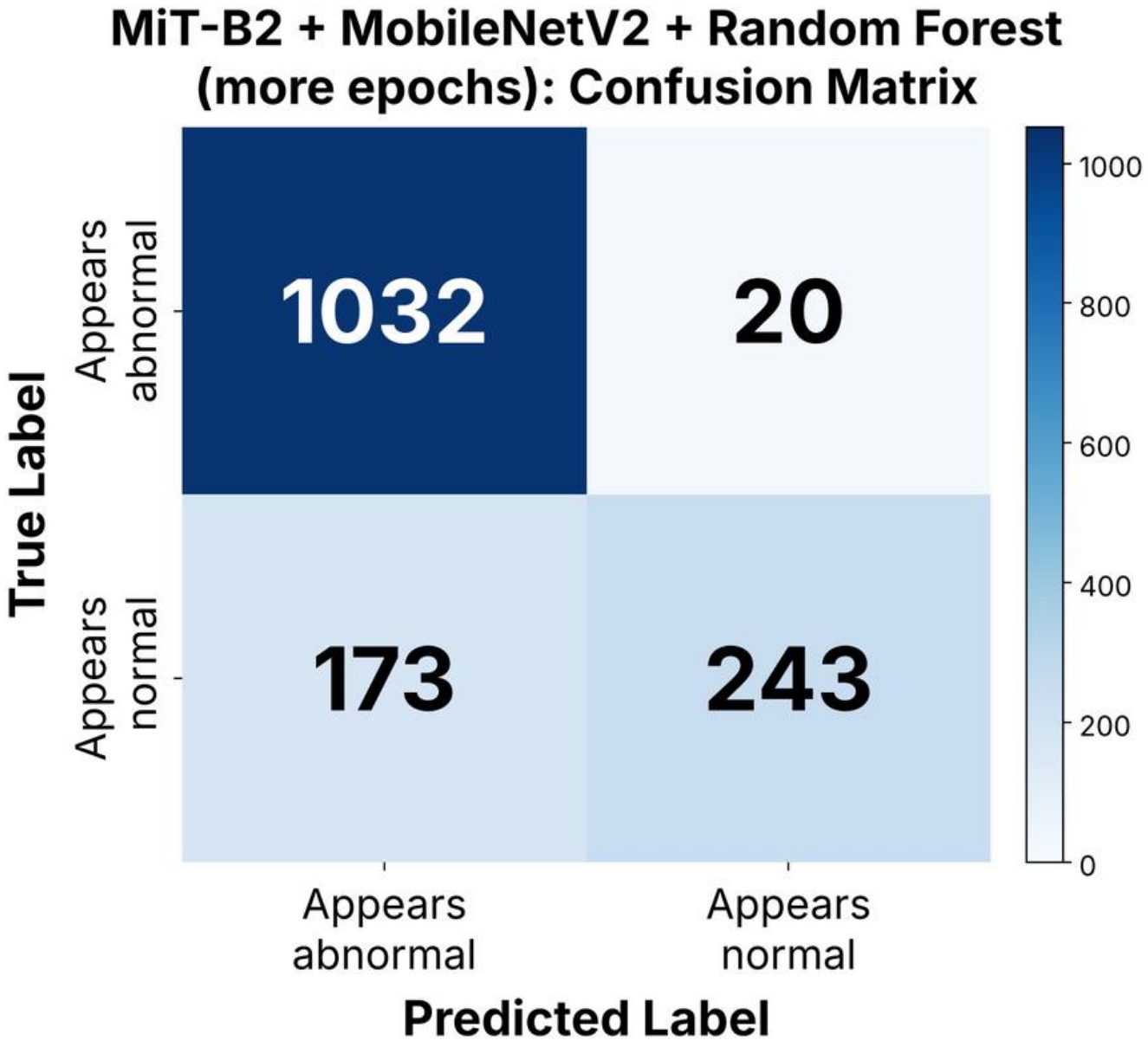
Accuracy: 86.85%

Precision: 87.56%

Recall: 86.85%

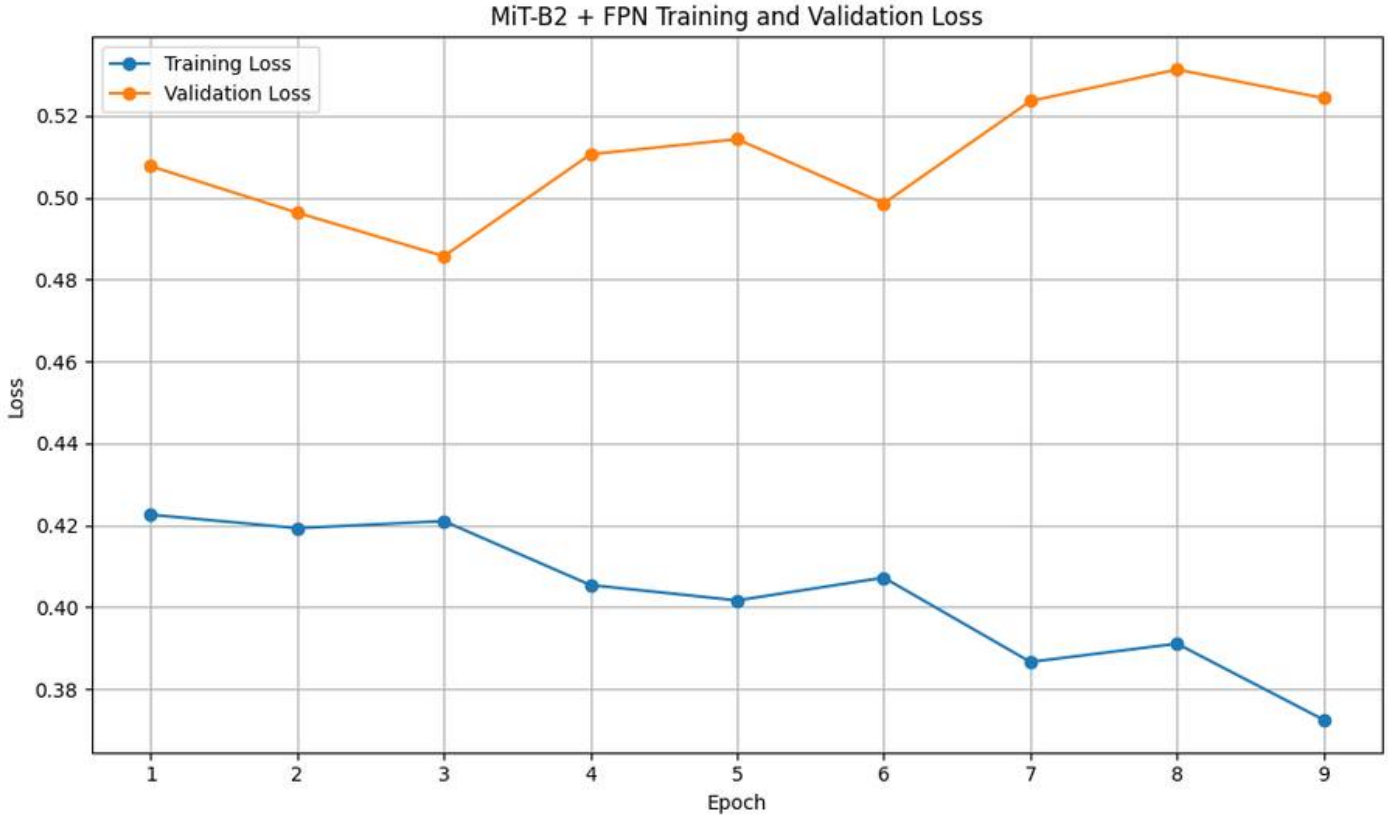
F1-score: 85.82%

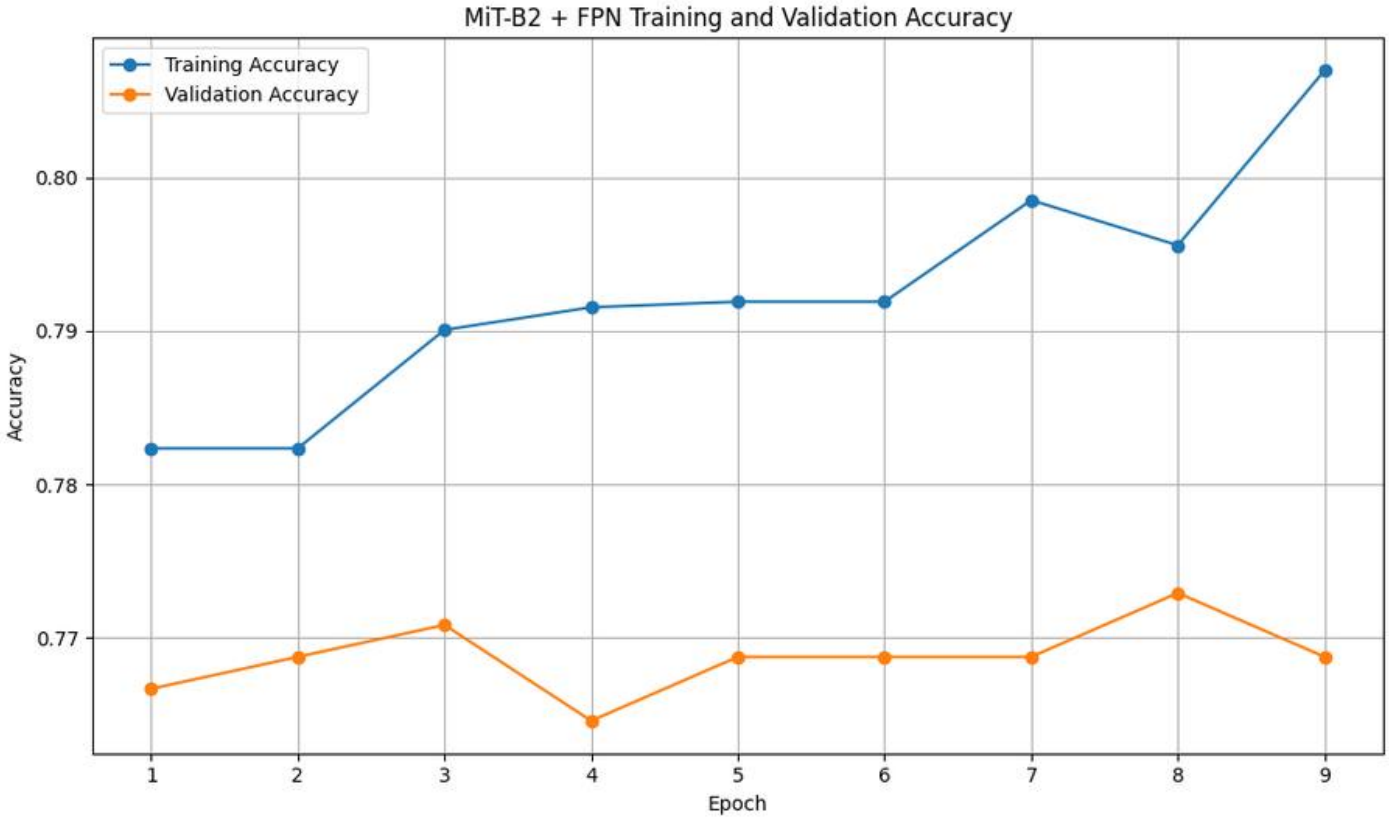
Confusion Matrix



8. MiT-B2 + MobileNetV2 + Random Forest (more epochs) — graphs (1/2)

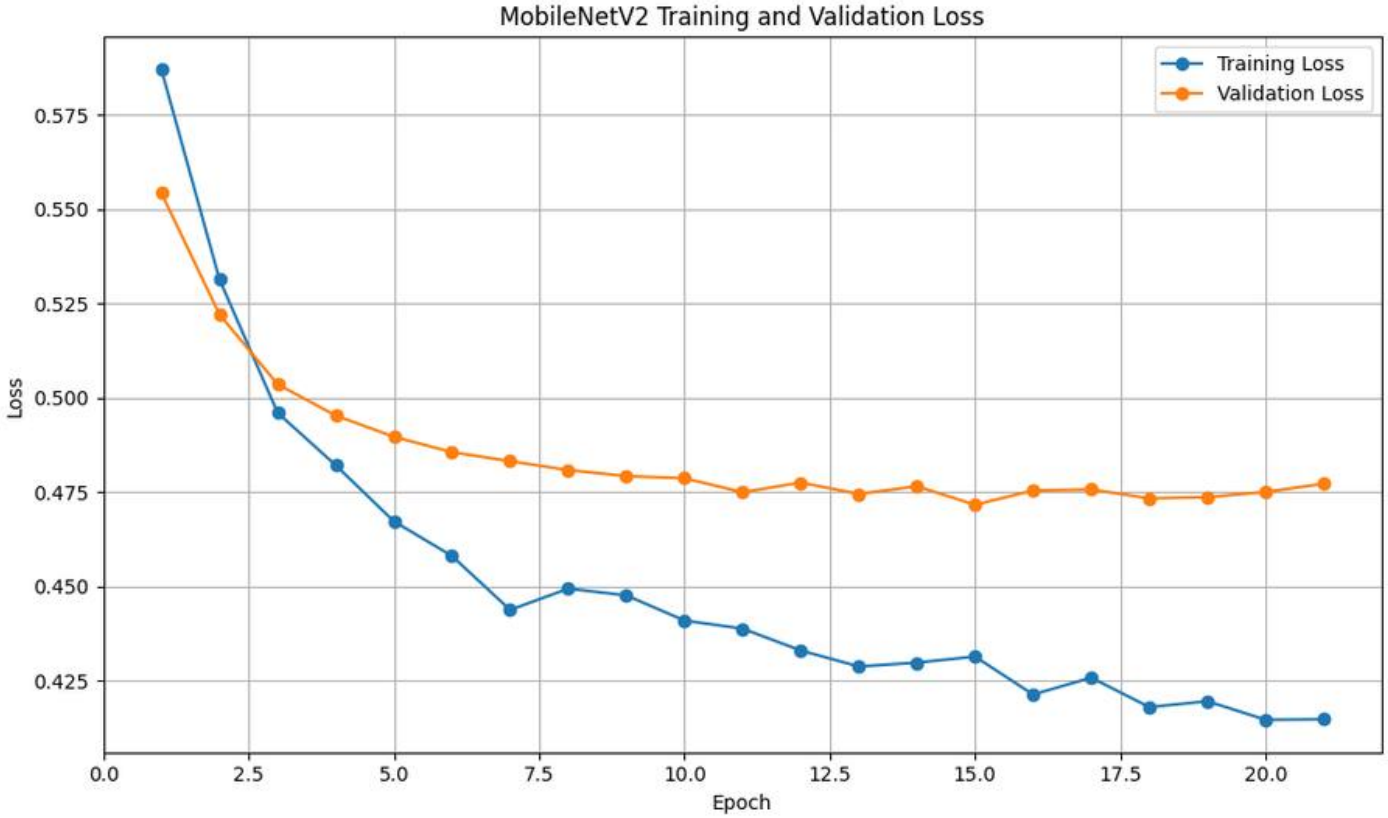
Training / Validation Graphs

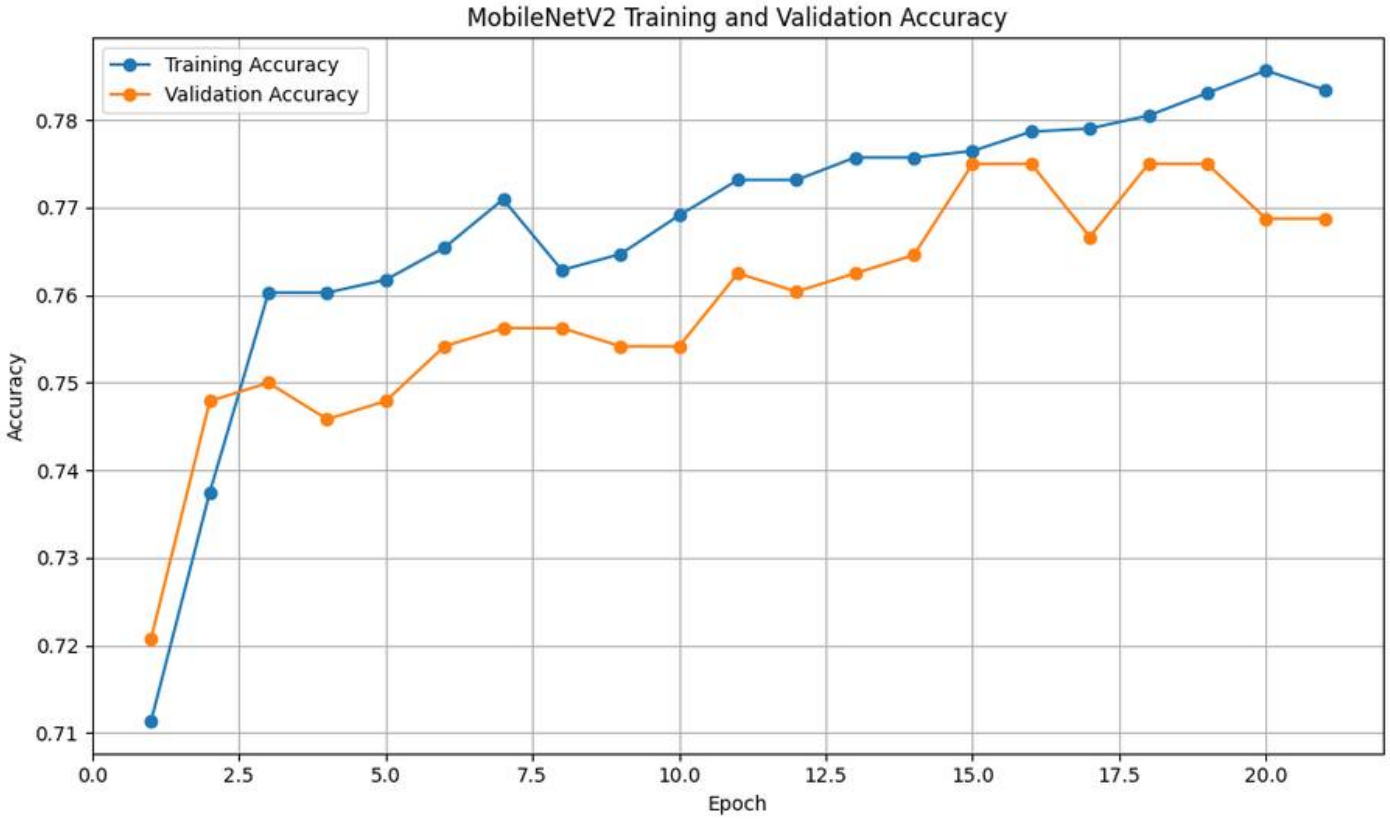




8. MiT-B2 + MobileNetV2 + Random Forest (more epochs) — graphs (2/2)

Training / Validation Graphs





9. MiT-B2 + MobileNetV2 + AdaBoost

Source notebook: MIT+ MobileNet+ ML.ipynb (inside MIT + MobileNet.zip)

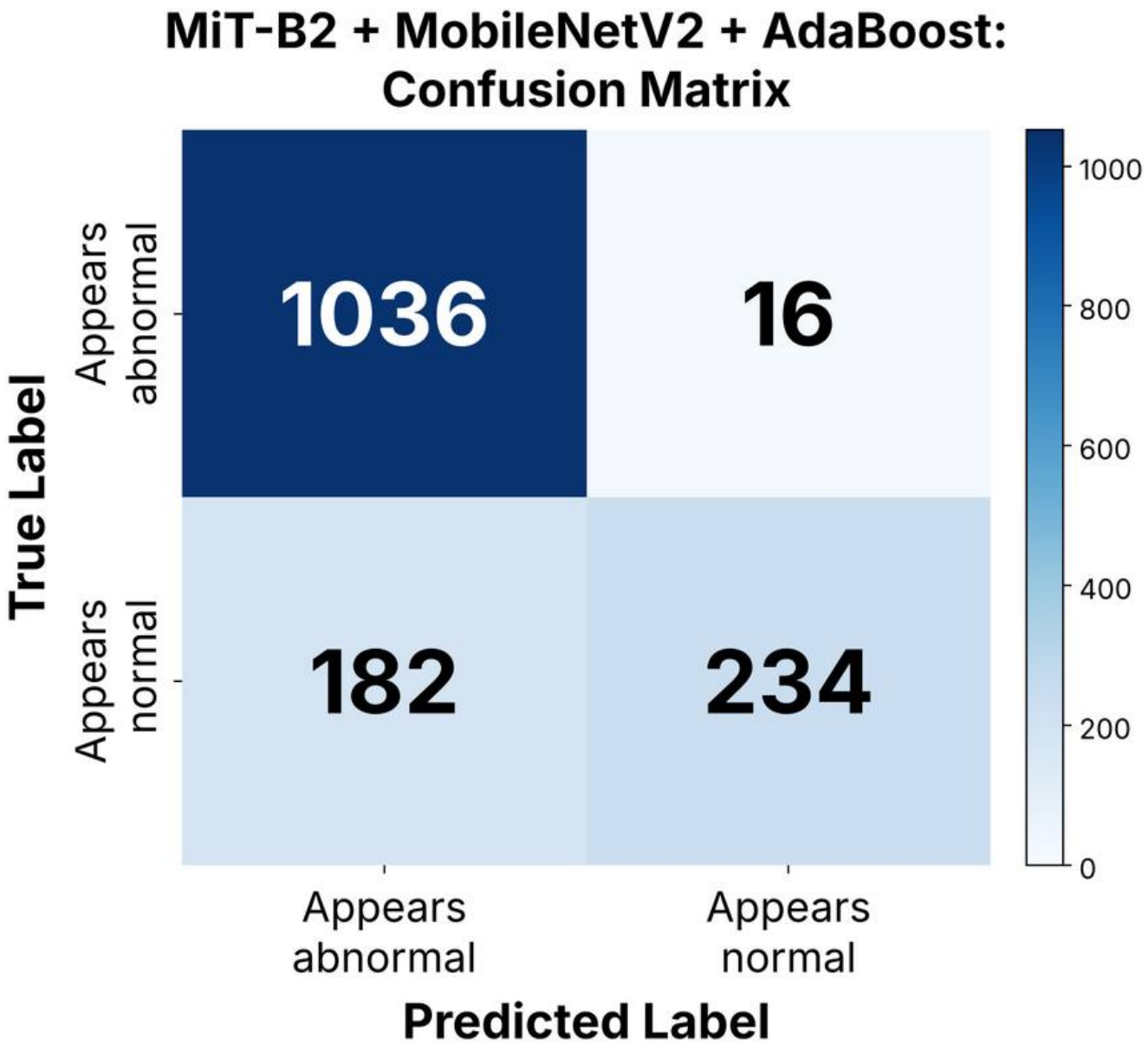
Accuracy: 86.51%

Precision: 87.48%

Recall: 86.51%

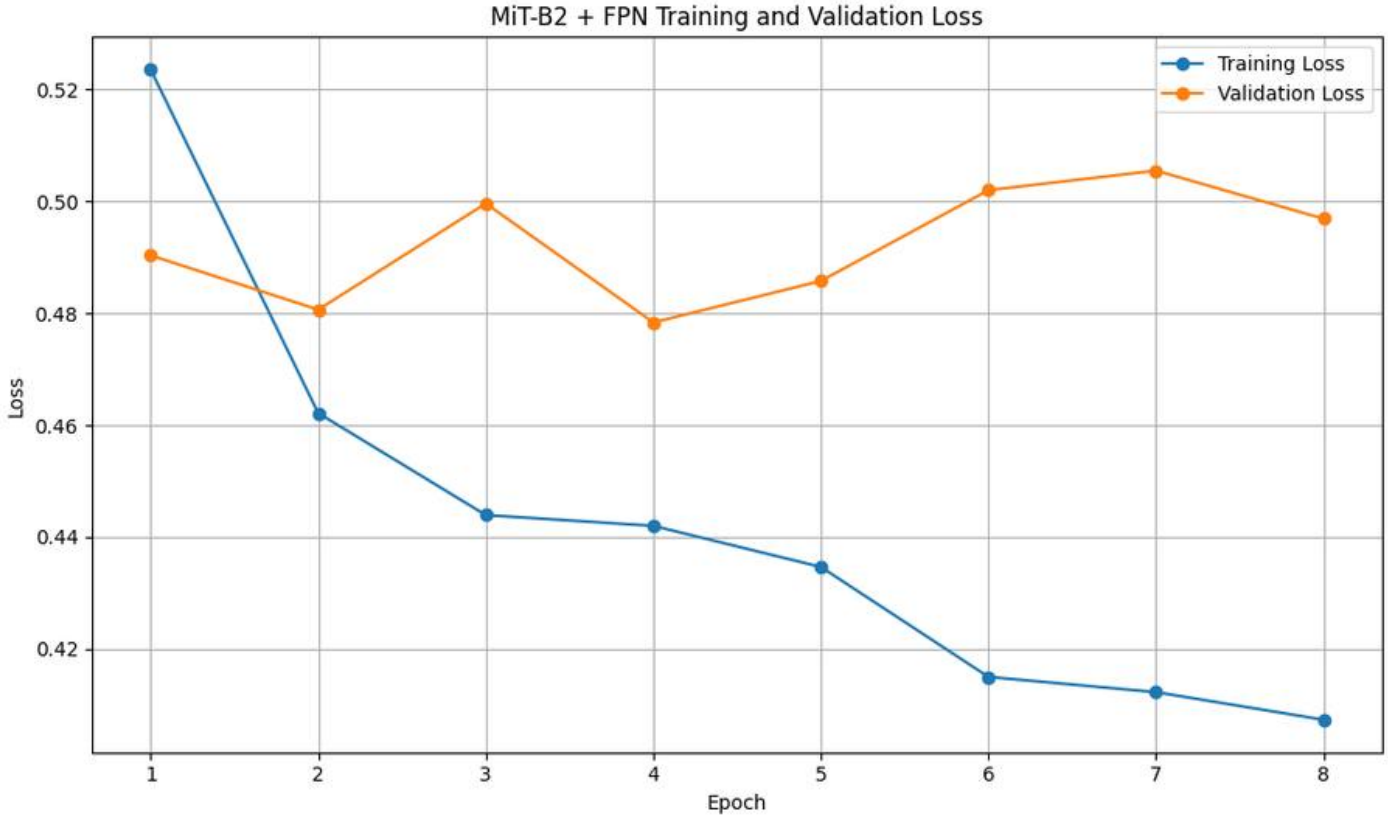
F1-score: 85.32%

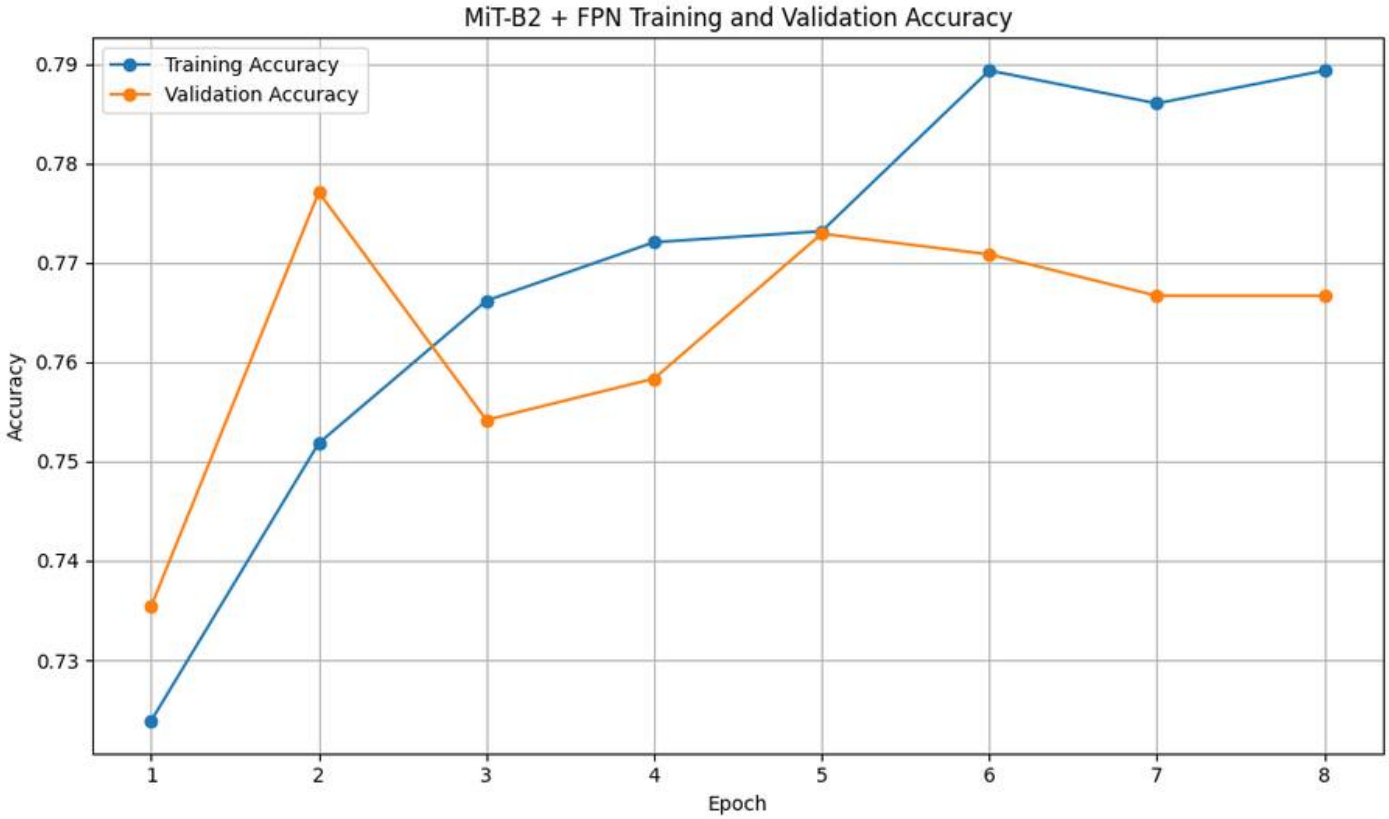
Confusion Matrix



9. MiT-B2 + MobileNetV2 + AdaBoost — graphs (1/2)

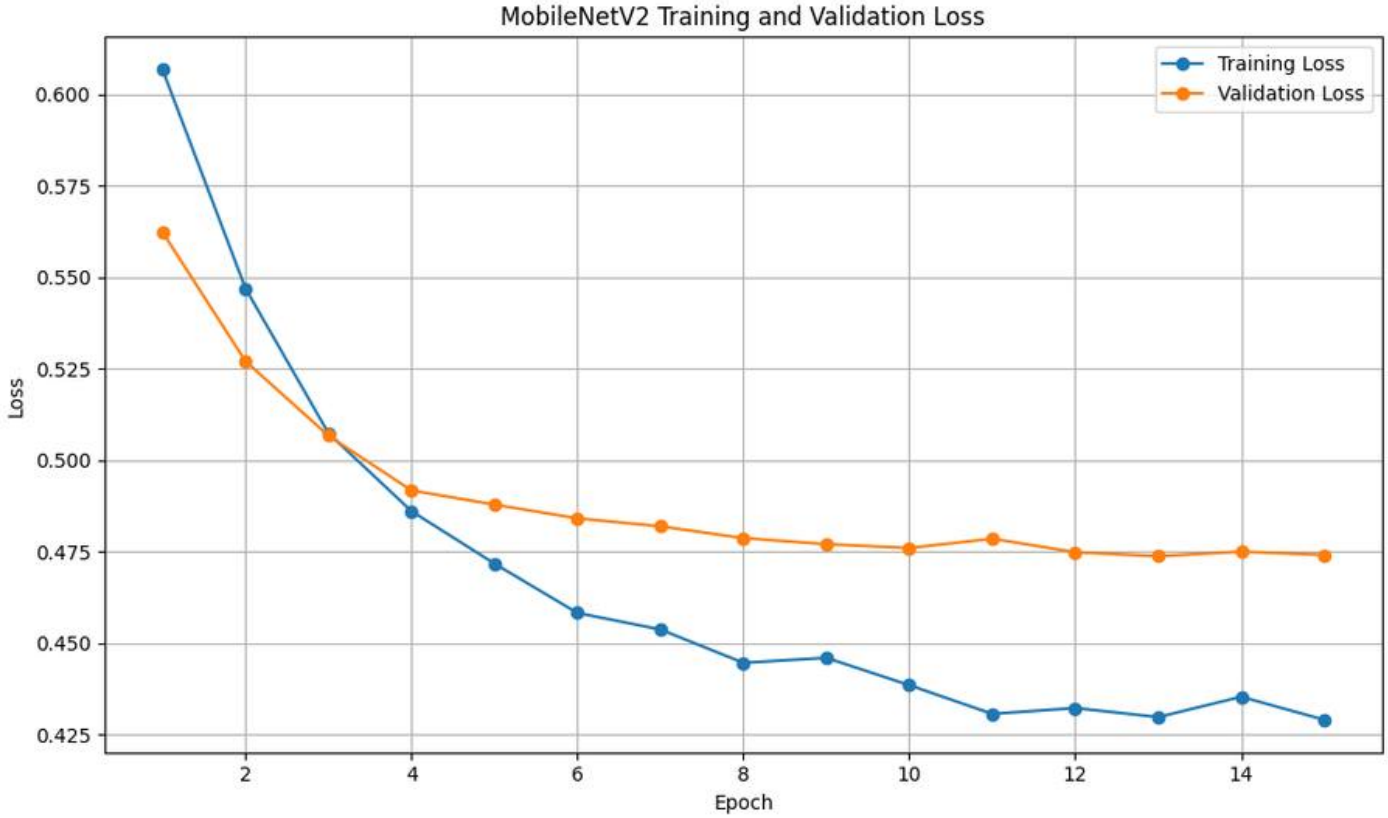
Training / Validation Graphs

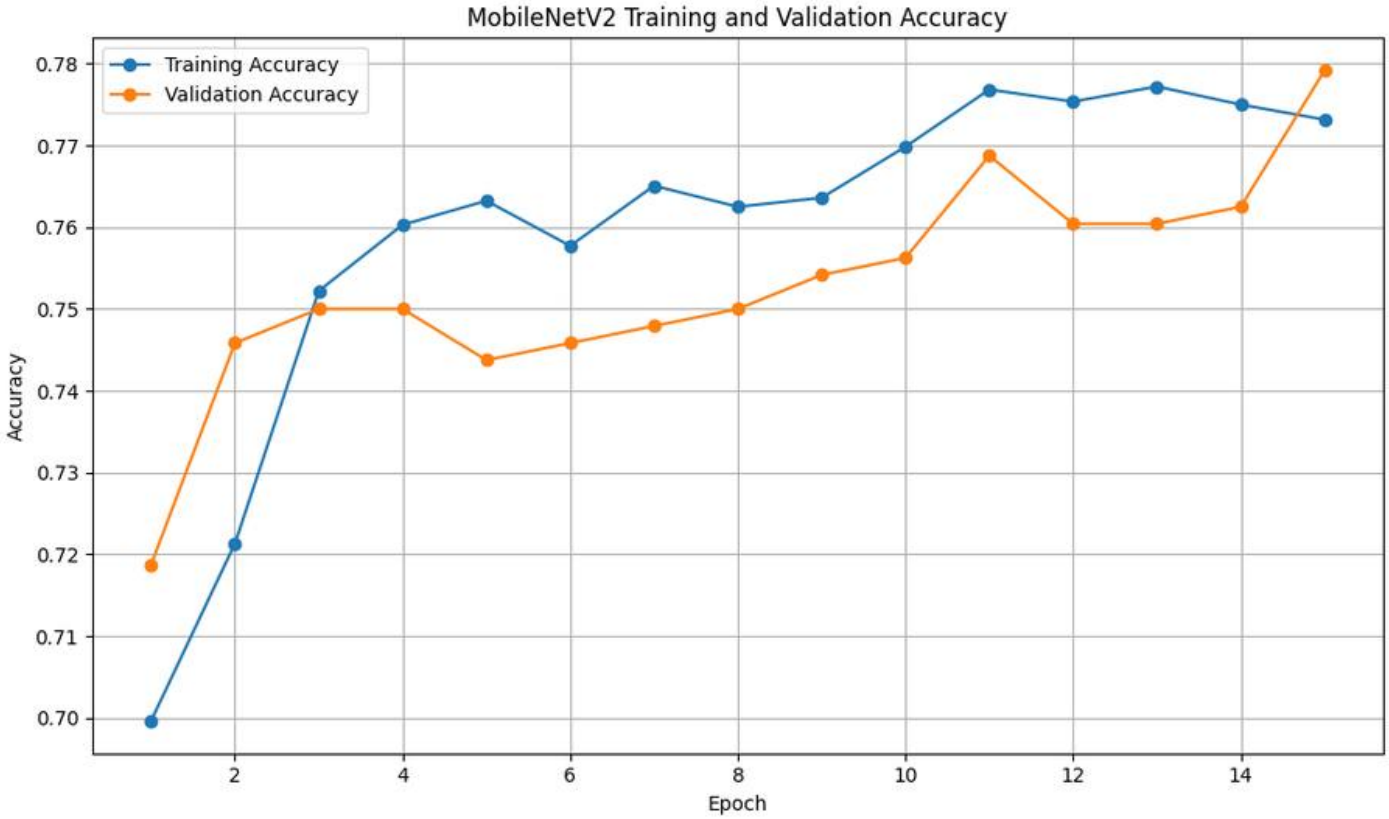




9. MiT-B2 + MobileNetV2 + AdaBoost — graphs (2/2)

Training / Validation Graphs





10. MiT-B2 + MobileNetV2 + AdaBoost (more epochs)

Source notebook: MIT more epochs.ipynb (inside MIT + MobileNet.zip)

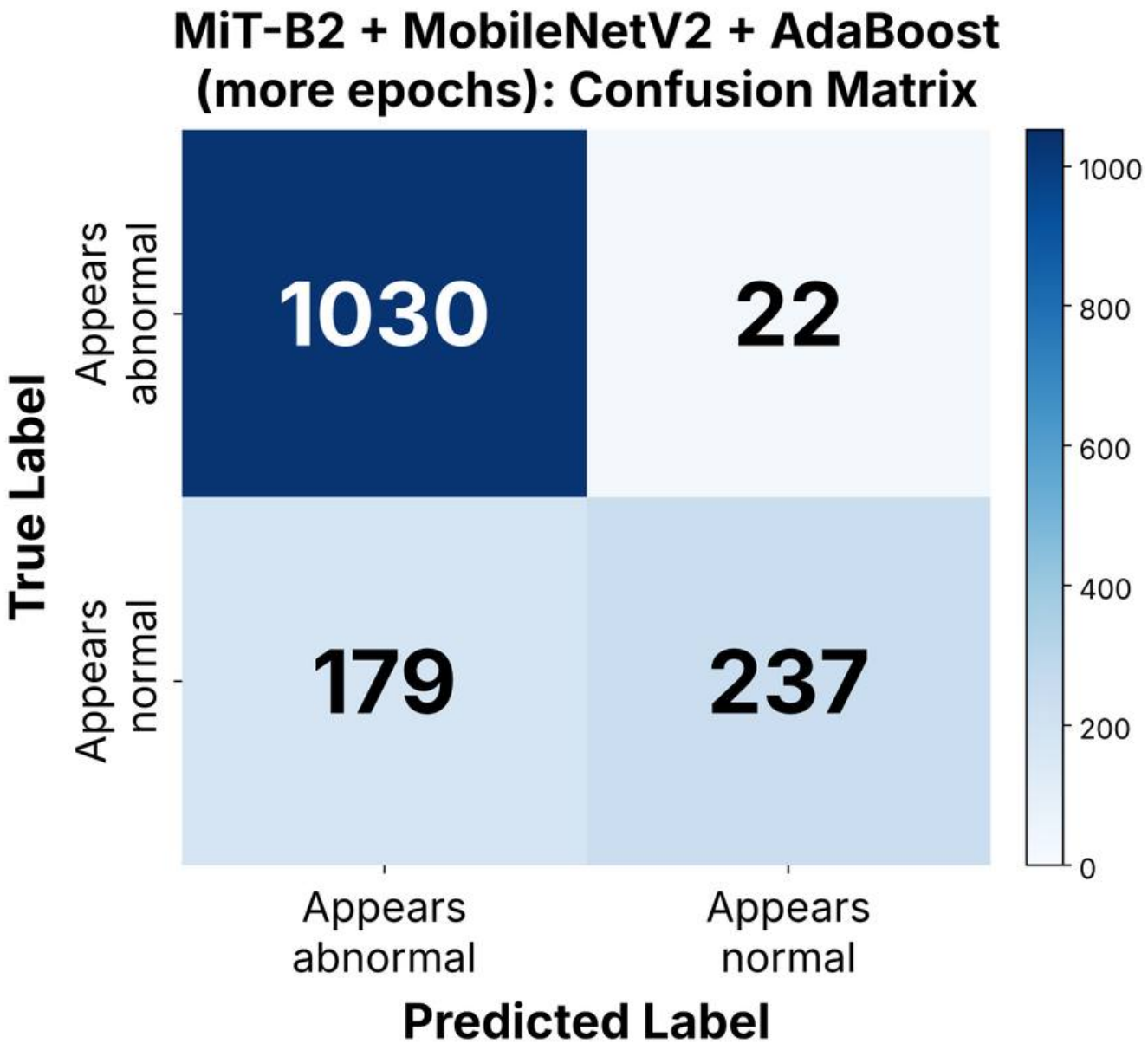
Accuracy: 86.31%

Precision: 86.98%

Recall: 86.31%

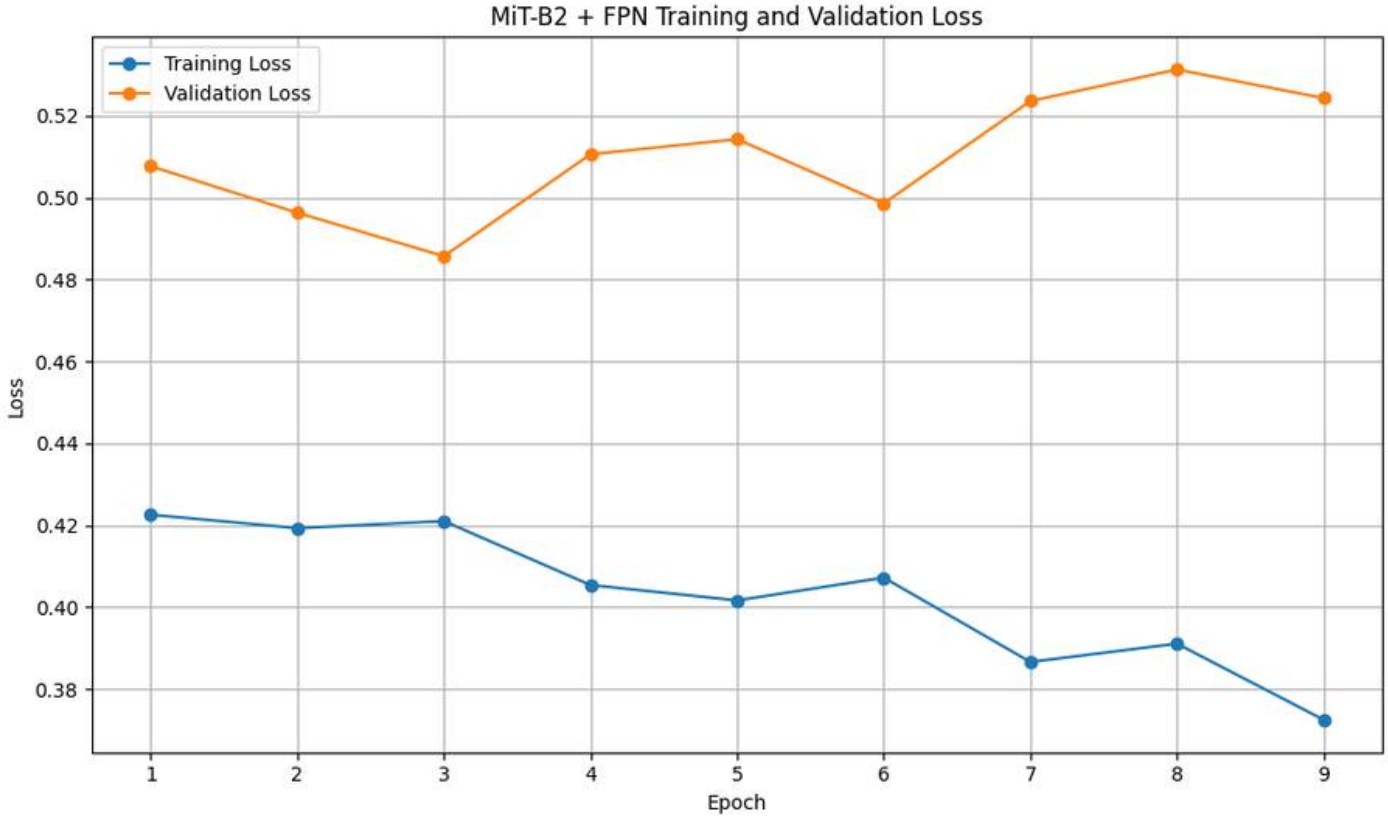
F1-score: 85.19%

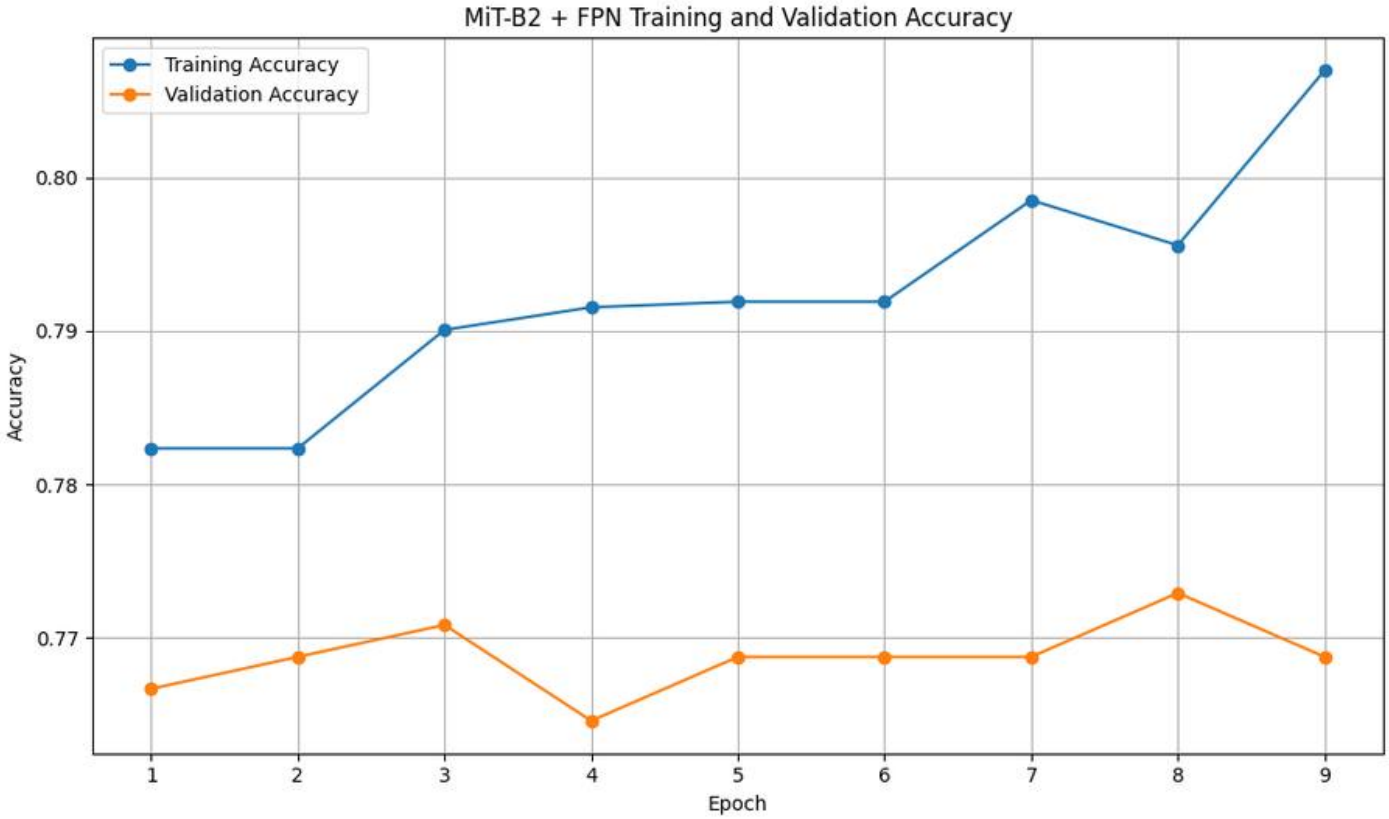
Confusion Matrix



10. MiT-B2 + MobileNetV2 + AdaBoost (more epochs) — graphs (1/2)

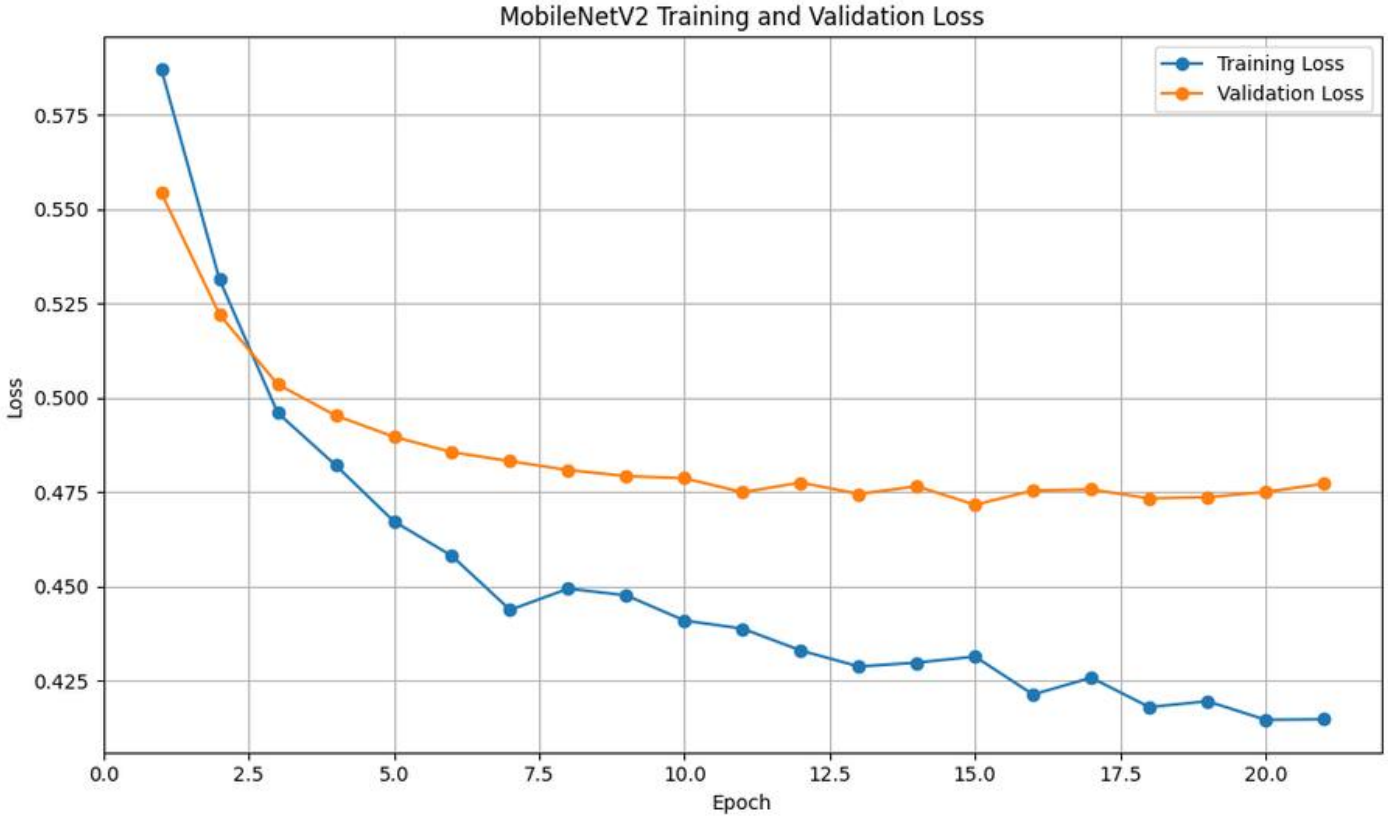
Training / Validation Graphs

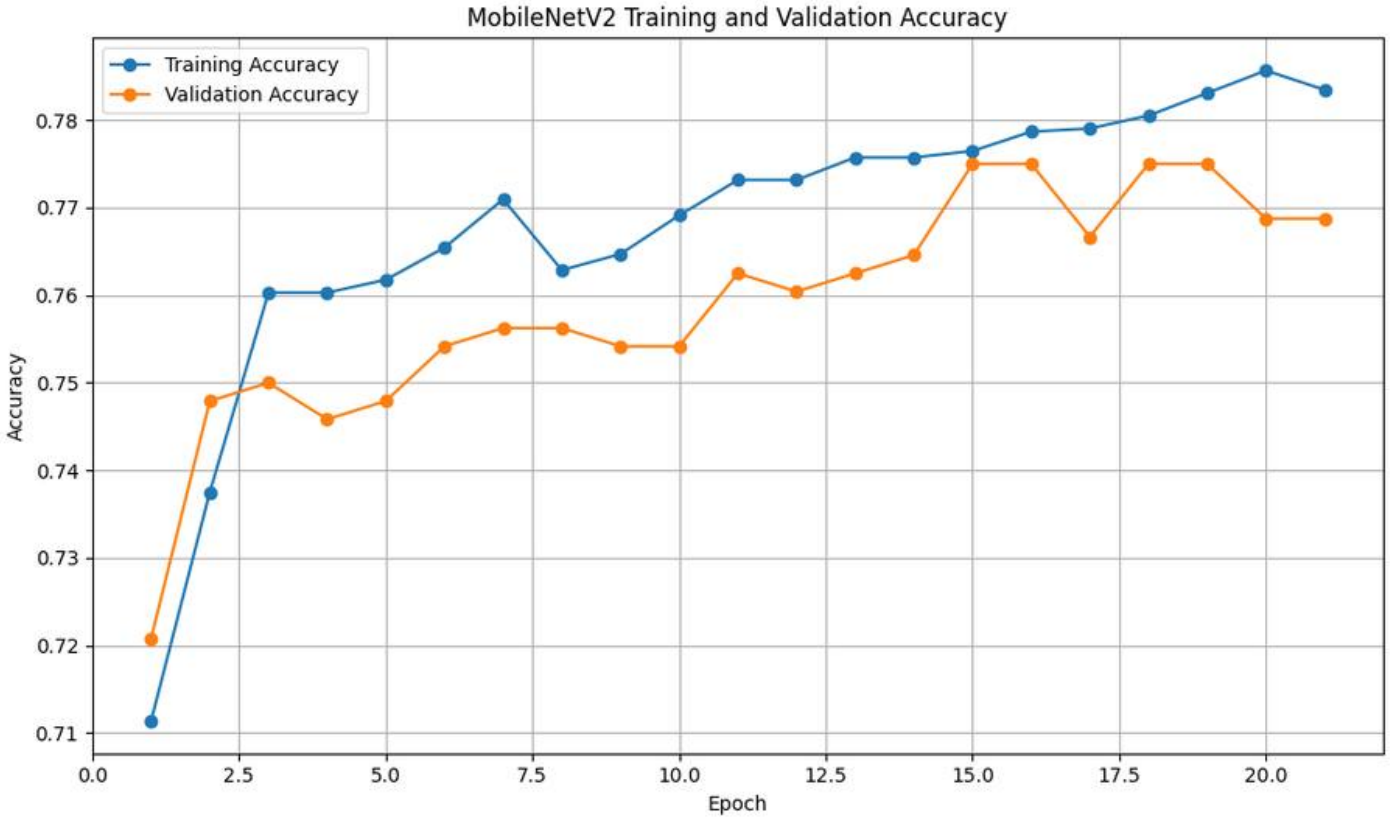




10. MiT-B2 + MobileNetV2 + AdaBoost (more epochs) — graphs (2/2)

Training / Validation Graphs





11. MiT-B2 + FPN baseline — Direct Classification

Source notebook: MIT + RF and MIT only.ipynb (inside MIT + MobileNet.zip)

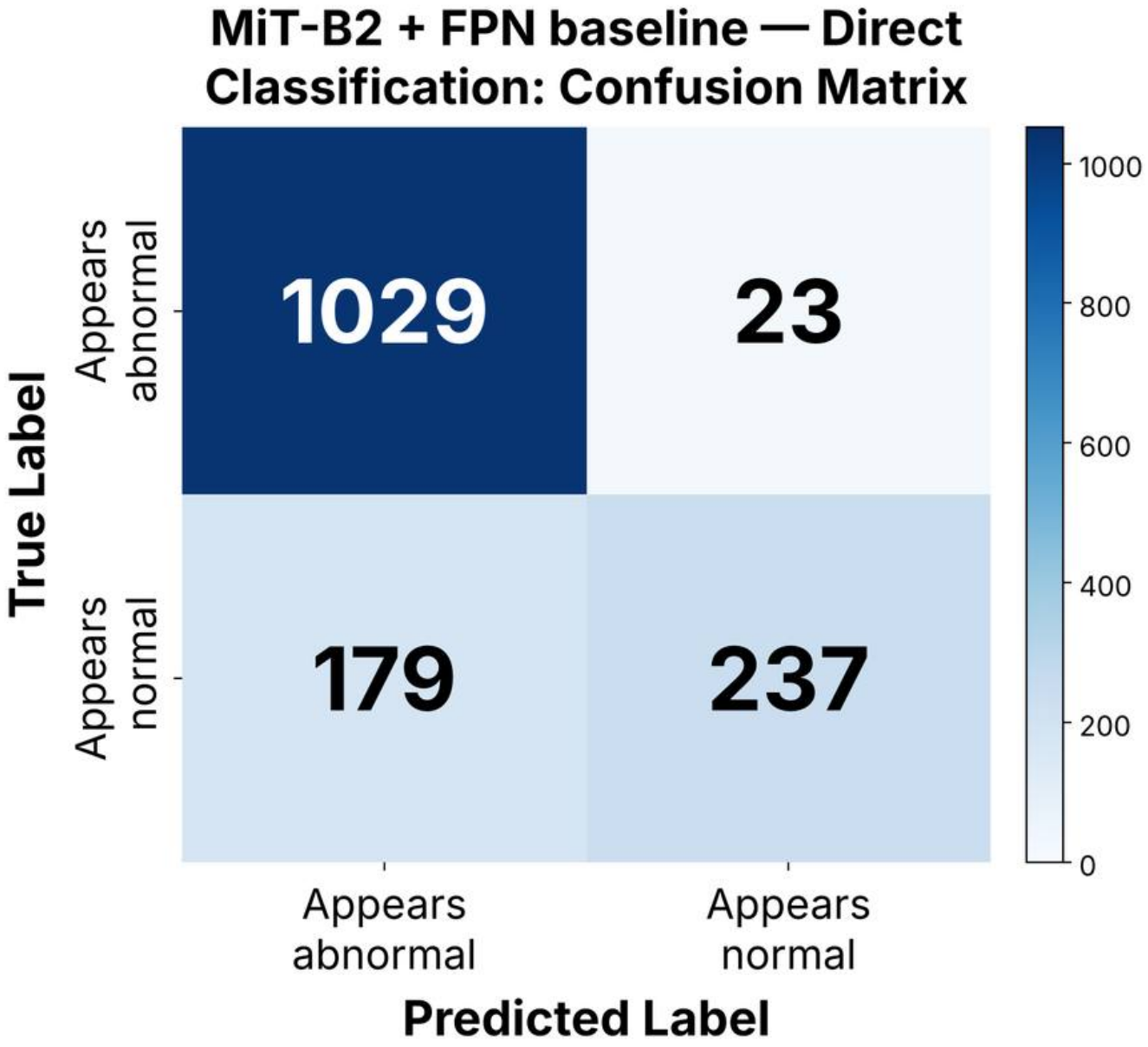
Accuracy: 86.24%

Precision: 86.87%

Recall: 86.24%

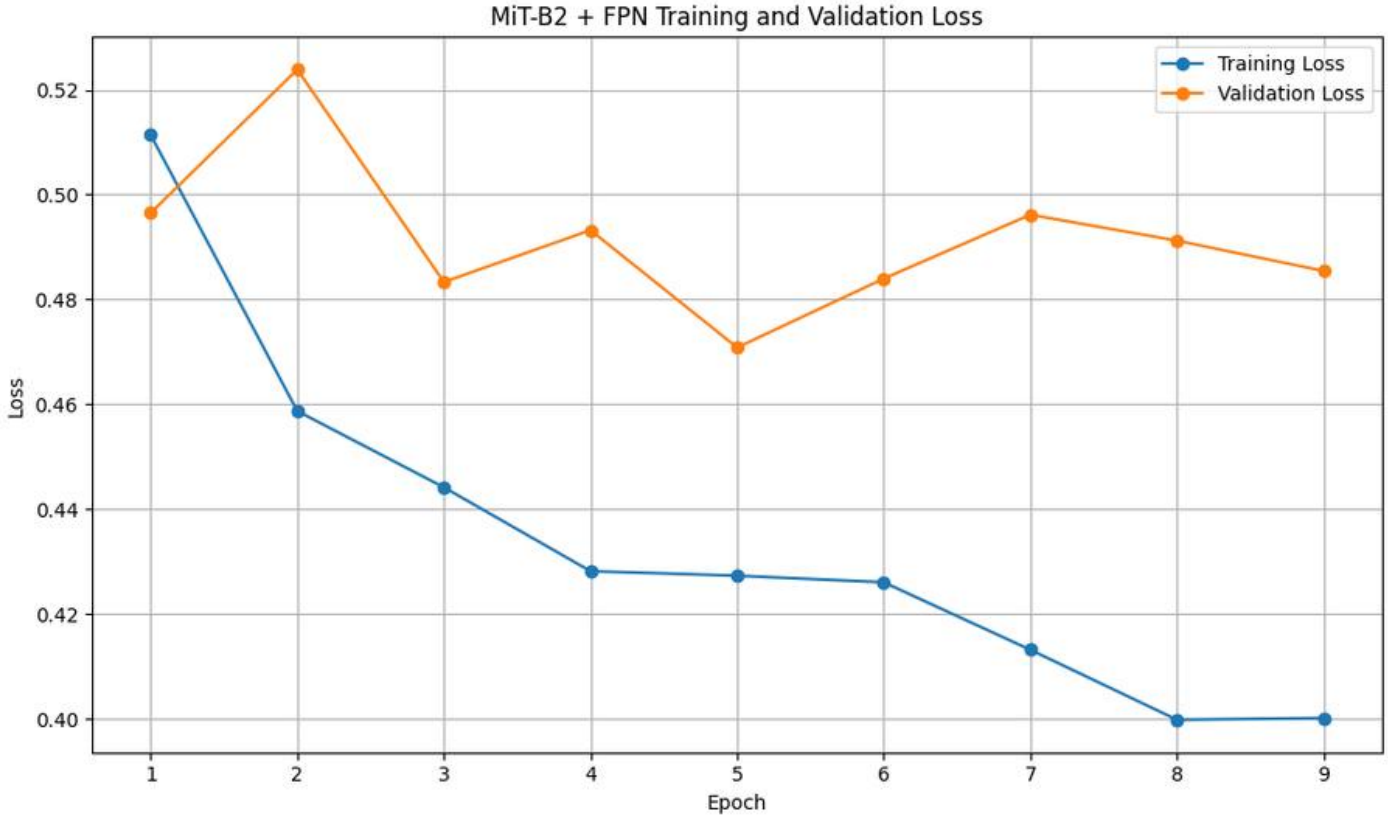
F1-score: 85.13%

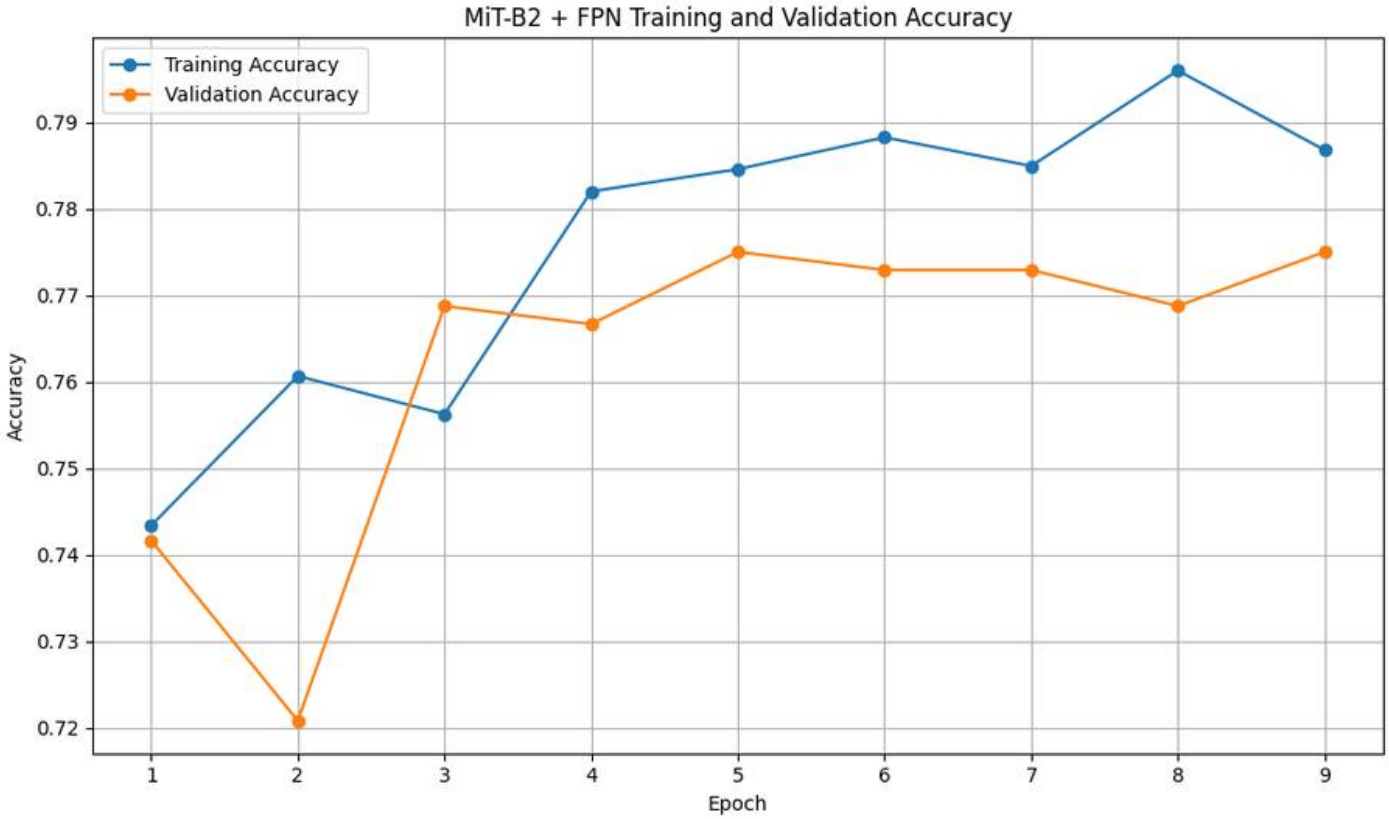
Confusion Matrix



11. MiT-B2 + FPN baseline — Direct Classification — graphs

Training / Validation Graphs





3. Notes

- Sources: the notebooks of MIT__MobileNet.zip (4 notebooks). Every value was read from the notebooks' saved outputs; nothing was re-run.
- Source column: "cell N" is the position of the cell inside the .ipynb file counted from 0 over all cells. Some notebooks also contain their own "# CELL n" comments, which use a different numbering and are not used here.
- Confusion matrices: copied from the notebook output where it prints a matrix; where only an image or per-class report exists, the counts were recovered exactly from per-class recall × support (1,052 / 416) and cross-checked with precision and accuracy. All matrices use the same colour map and the same fixed colour scale (0–1,052 images).
- Graphs: only values printed in the notebooks' training logs are plotted — nothing is estimated, smoothed or interpolated. Where the notebook printed a complete epoch log the graph is redrawn from it (with Train/Val CE curves added); the notebook's own original figures are shown wherever they exist. "CE" is torch.nn.CrossEntropyLoss; the combined loss is 0.5 × CE + 0.5 × Dice loss. In all scripts the validation images carry the training augmentations, so validation accuracy is lower and noisier than test accuracy.
- Where a notebook prints a test accuracy that disagrees with its own confusion matrix and weighted recall (Vgg kfold prints 81.06% against a recall of 84.20%; Vgg with Dice Loss prints 88.84%, which is the best validation accuracy), the table uses the accuracy implied by the confusion matrix, which equals the weighted recall. The page of such a row says so.
- The MiT-B2 notebooks fix seed 42 but contain no repeat run or saved artefacts, so they are not verified as reproducible.
- The MiT-B2 + FPN and MobileNetV2 trainings in these notebooks used plain CrossEntropyLoss only (no Dice term, no combined loss); their graphs are the notebooks' original figures.
- "more epochs", "Tuned" and the other labels in the run names are taken from the notebook titles.